

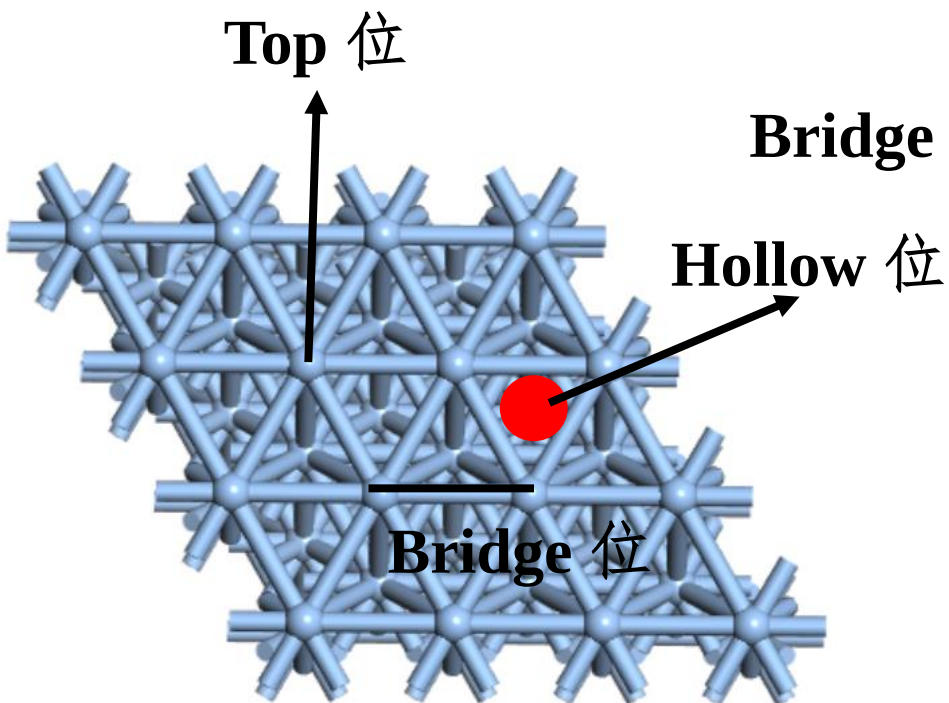
# 电催化计算实战入门

## 热力学计算

主讲人：高怡静

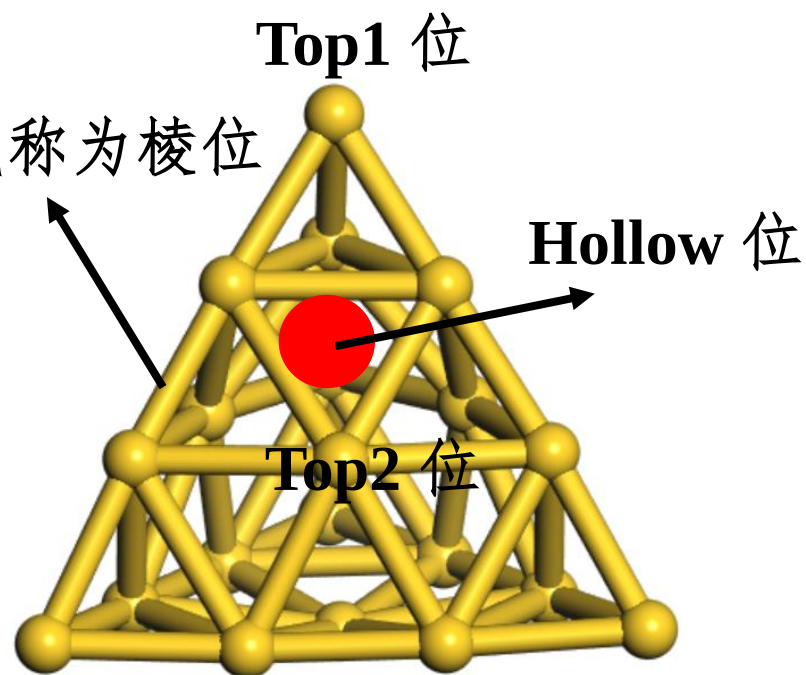
# 活性位点的判断

金属平面活性位点判断：  
以Ag(111)面为例



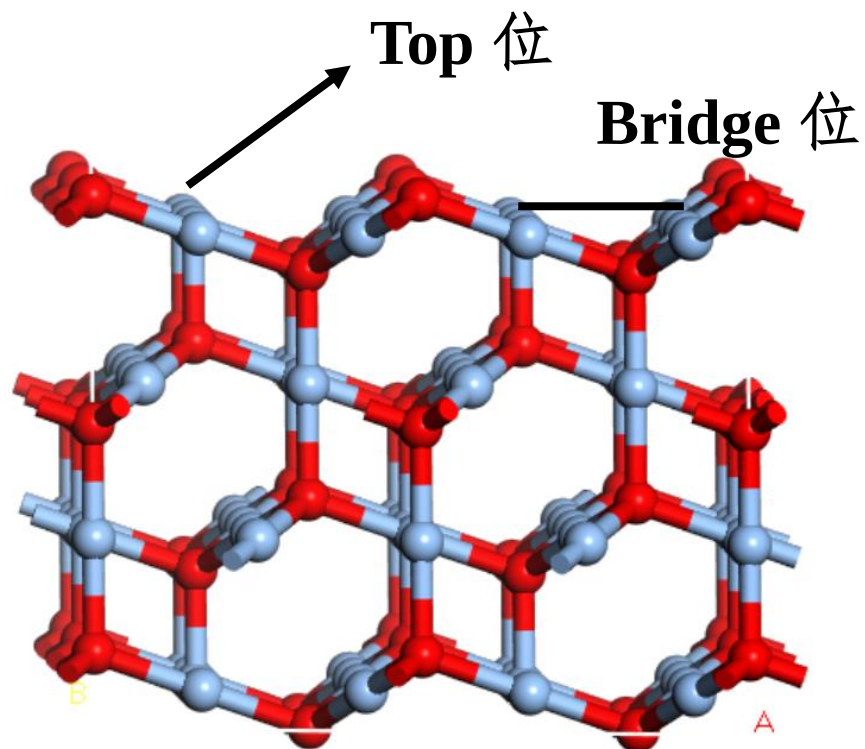
Ag(111)面俯视图

团簇活性位点判断：  
以Au<sub>20</sub>面为例

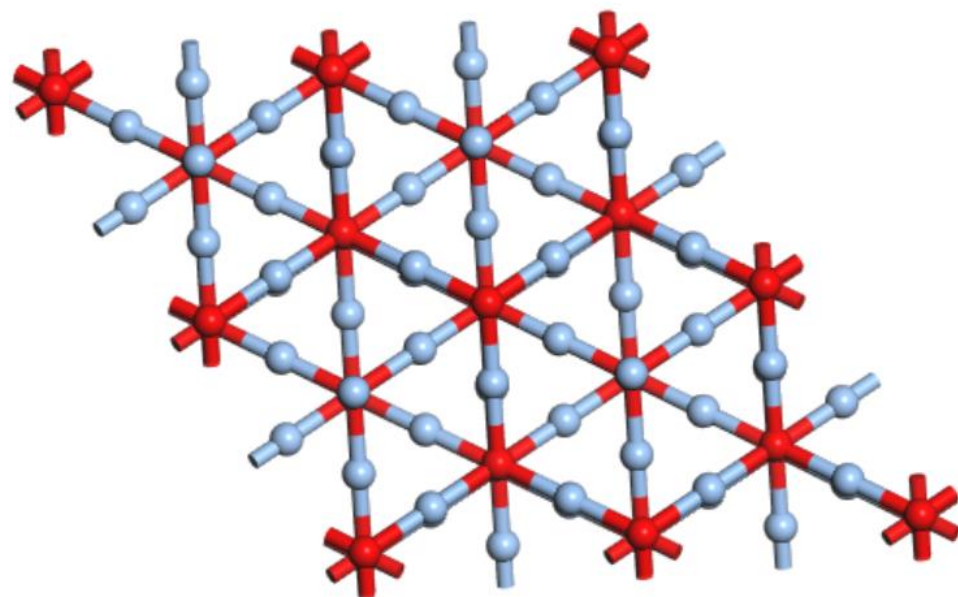


Au<sub>20</sub>结构图

金属氧化物活性位点判断：以 $\text{Ag}_2\text{O}(111)$ 面为例



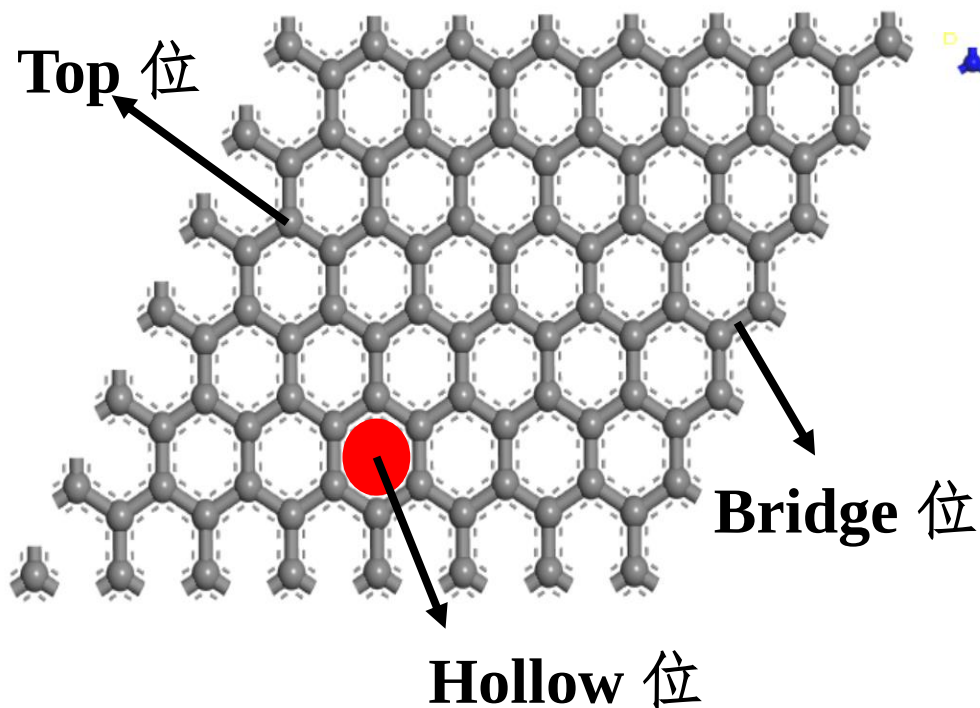
$\text{Ag}_2\text{O}(111)$ 面主视图



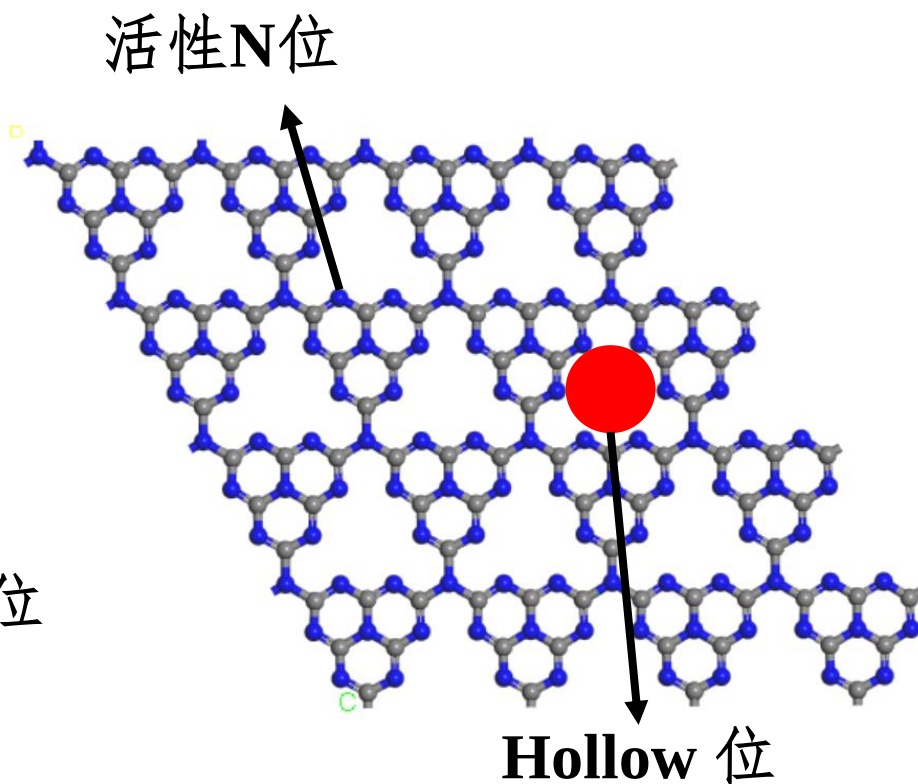
$\text{Ag}_2\text{O}(111)$ 面俯视图

# 活性位点的判断

## 二维材料活性位点判断:

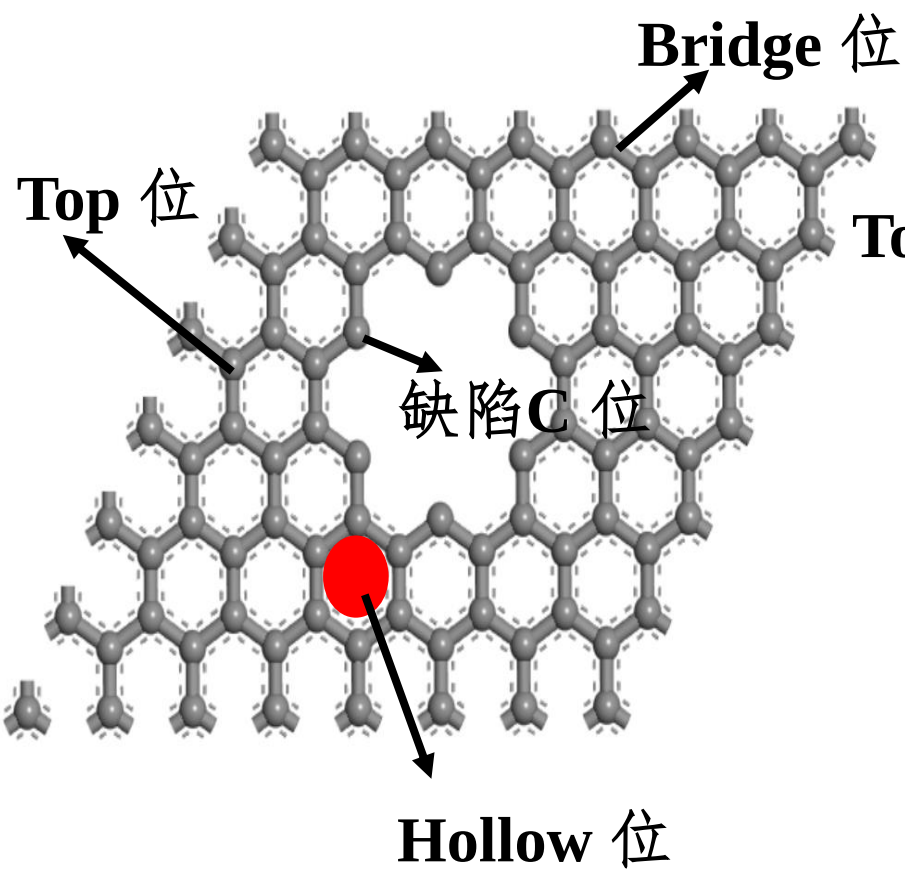


石墨烯

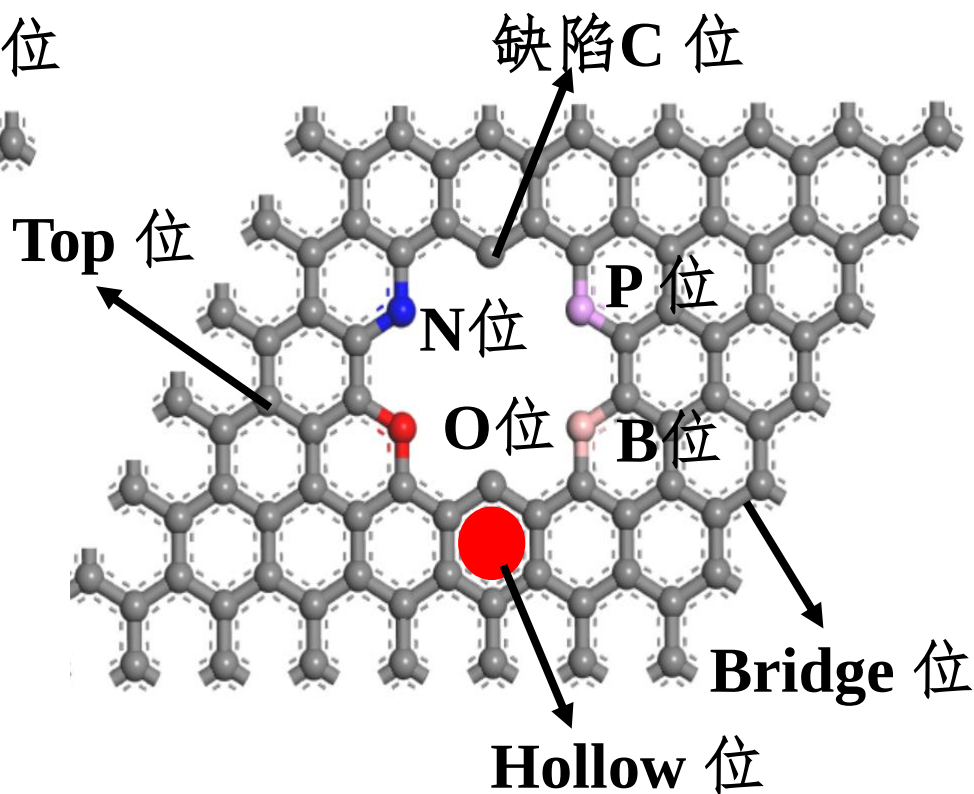


g-C<sub>3</sub>N<sub>4</sub>

## 二维材料活性位点判断：



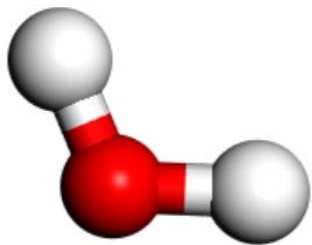
缺陷石墨烯



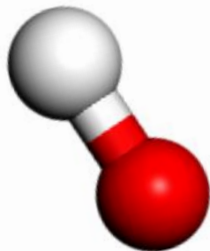
B,N,O,P掺杂的原子石墨烯



**HER**计算涉及到的关键中间体



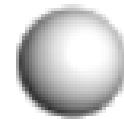
$\text{H}_2\text{O}(\text{g})$



OH



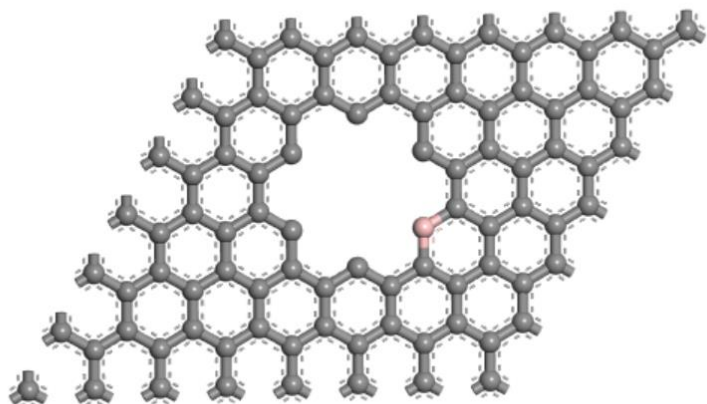
O



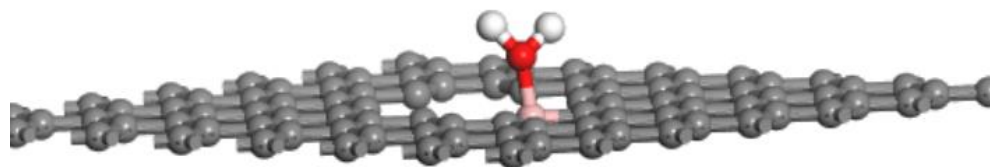
H

# HER关键中间体吸附模型构建

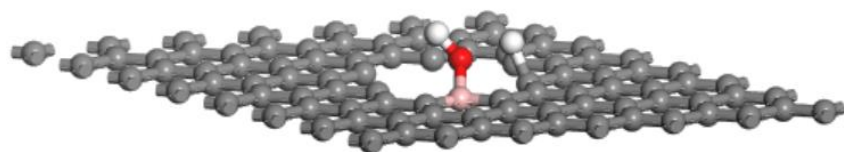
**HER**关键中间体吸附模型（以**B**掺杂缺陷石墨烯**HER**为例）



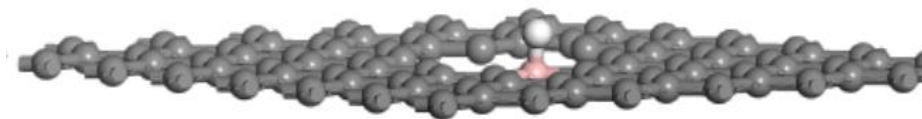
**B@V\_Graphene**



**\*H<sub>2</sub>O**

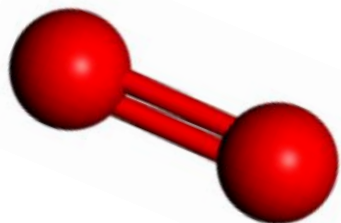


**\*OH + \*H**

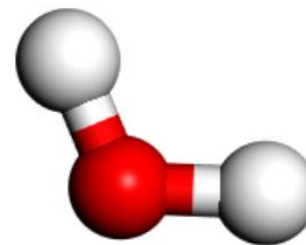


**\*H**

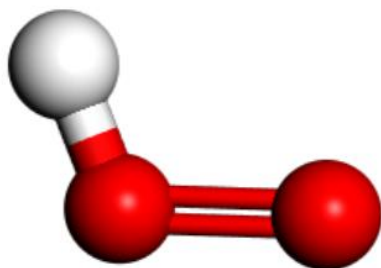
## ORR/OER计算涉及到的关键中间体



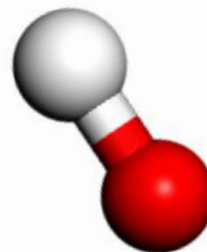
$\text{O}_2(\text{g})$



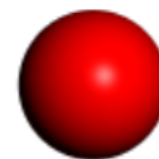
$\text{H}_2\text{O}(\text{g})$



$\text{OOH}$



$\text{OH}$

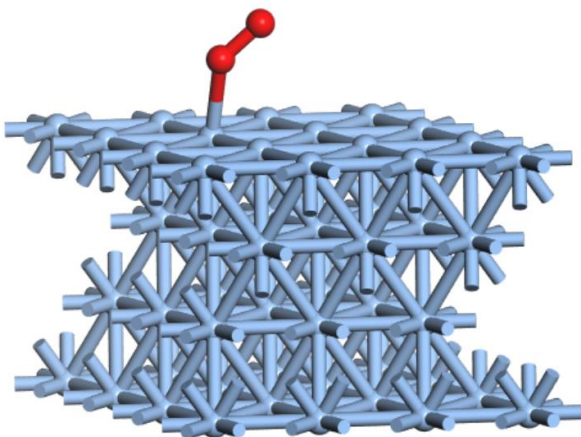


$\text{O}$

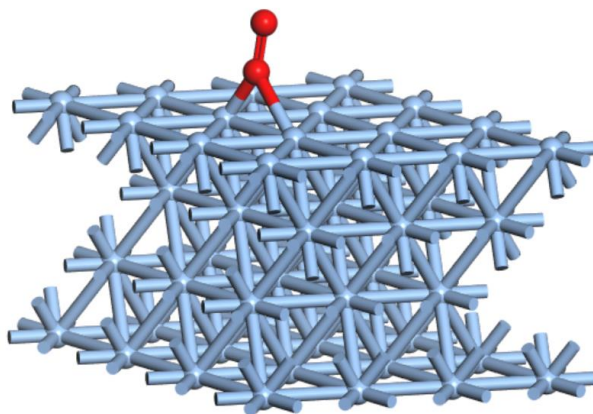


**ORR/OER**关键中间体吸附模型（以Ag(111)面的ORR/OER为例）

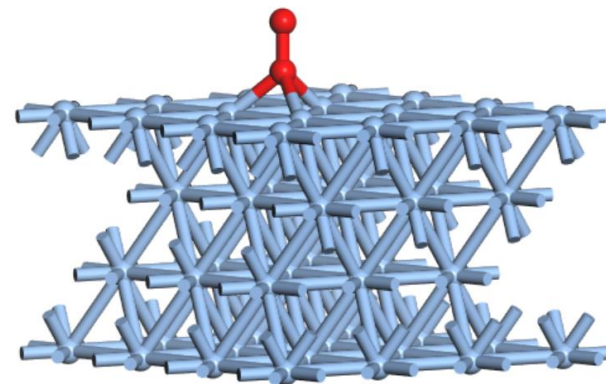
**O<sub>2</sub>**的吸附位



**TOP位**



**Bridge位**

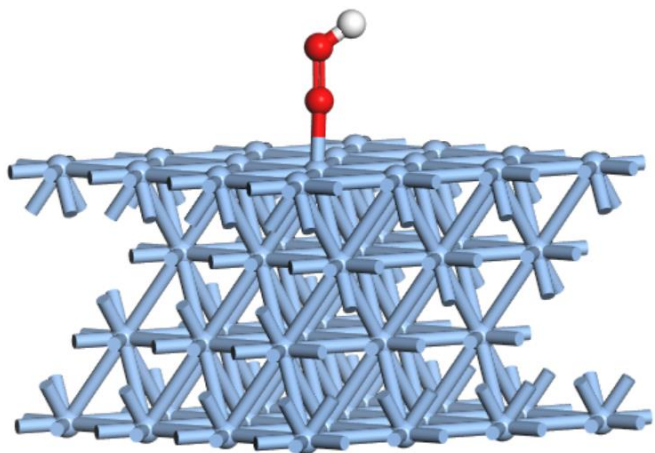


**Hollow位**

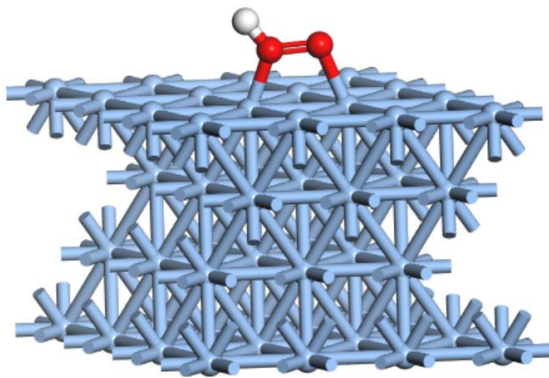
根据计算得到的能量，确定活性位点。

**ORR/OER**关键中间体吸附模型（以Ag(111)面的ORR/OER为例）

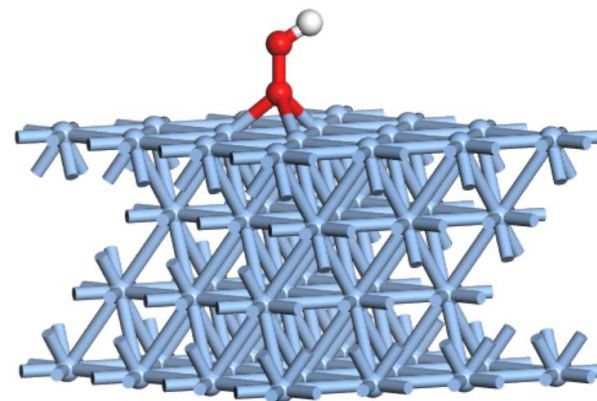
**OOH**的吸附位



**TOP**位



**Bridge**位

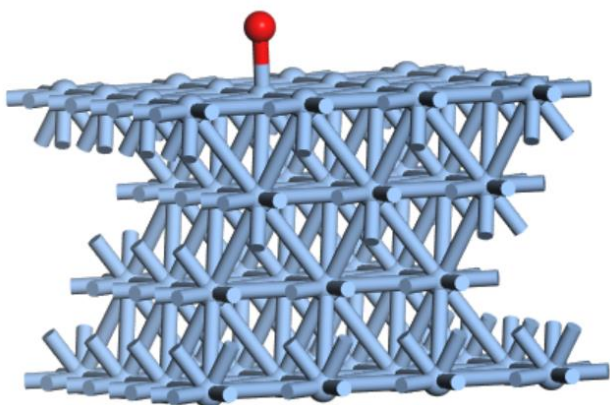


**Hollow**位

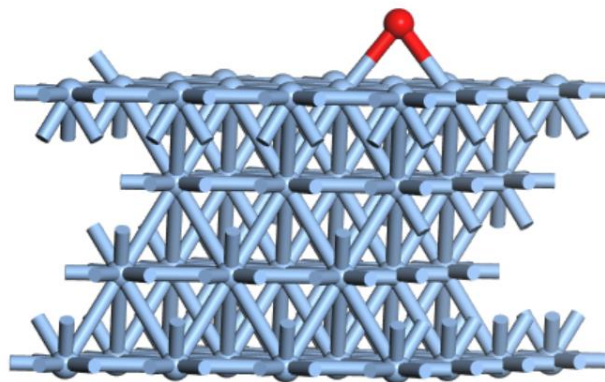
根据计算得到的能量，确定活性位点。

## ORR/OER关键中间体吸附模型（以Ag(111)面的ORR/OER为例）

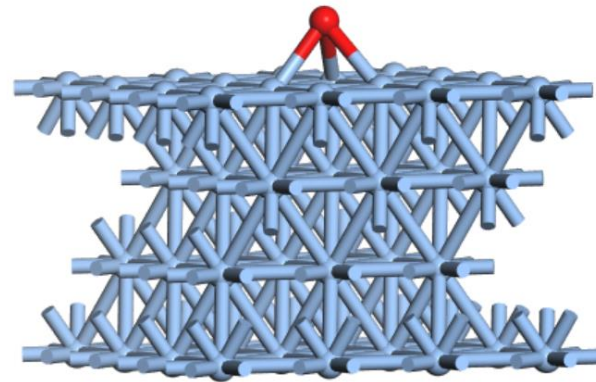
O的吸附位



TOP位



Bridge位



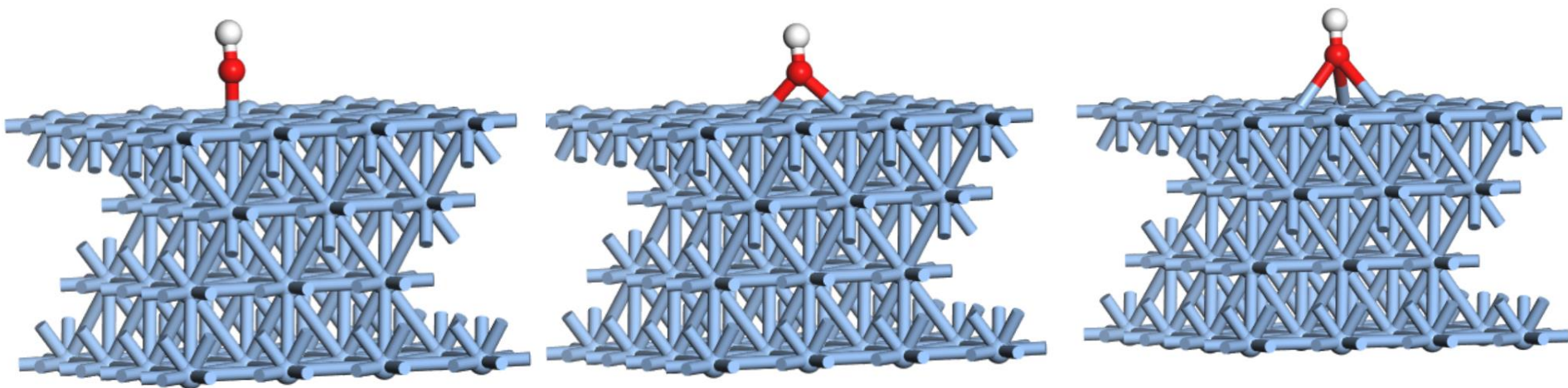
Hollow位

根据计算得到的能量，确定活性位点。



## ORR/OER关键中间体吸附模型（以Ag(111)面的ORR/OER为例）

### OH的吸附位



TOP位

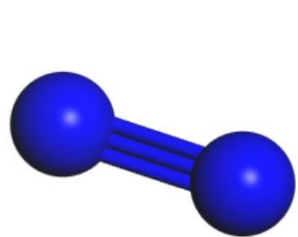
Bridge位

Hollow位

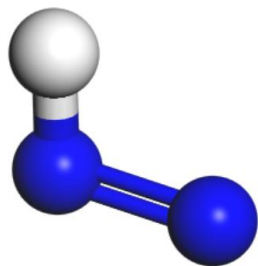
根据计算得到的能量，确定活性位点。

# NRR关键中间体的优化

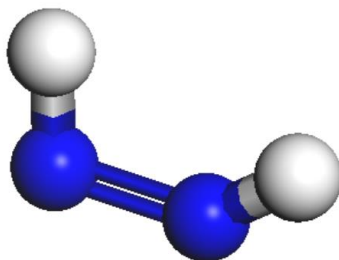
**NRR**计算涉及到的关键中间体



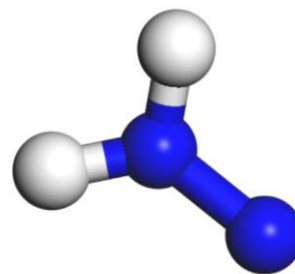
$\text{N}_2(\text{g})$



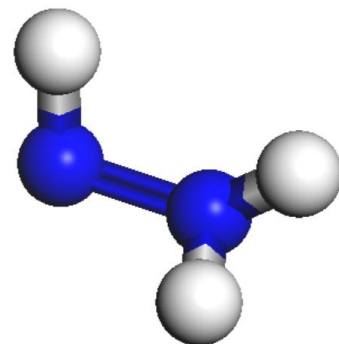
$\text{NNH}$



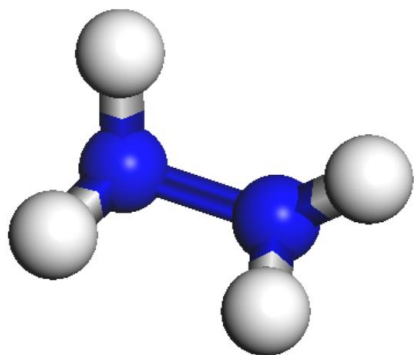
$\text{NHNH}$



$\text{NNH}_2$



$\text{NHNH}_2$



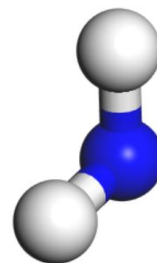
$\text{NH}_2\text{NH}_2$



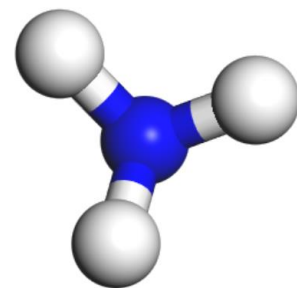
$\text{N}$



$\text{NH}$

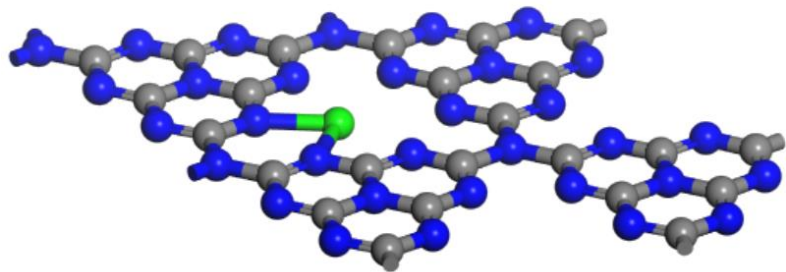


$\text{NH}_2$

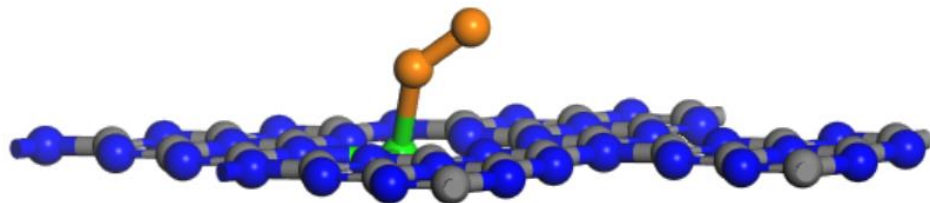


$\text{NH}_3(\text{g})$

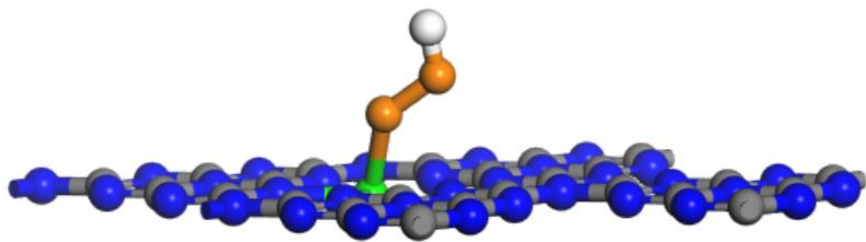
## NRR关键中间体吸附模型（以Pd@C<sub>3</sub>N<sub>4</sub>面的NRR为例）



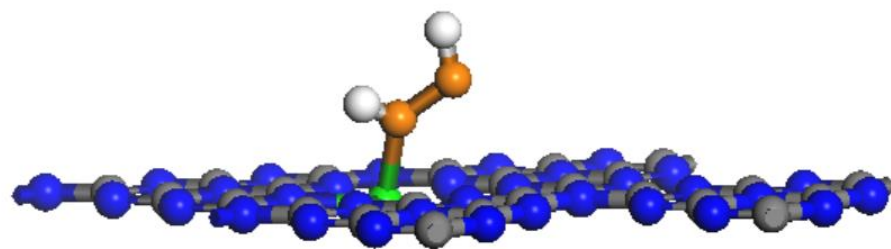
Pd@C<sub>3</sub>N<sub>4</sub>



\*N<sub>2</sub>



\*NNH

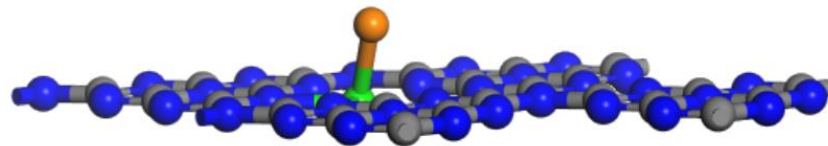
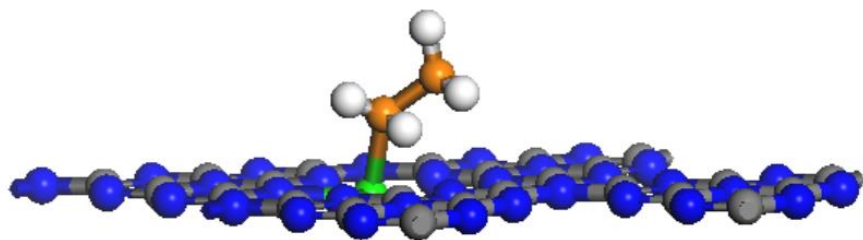
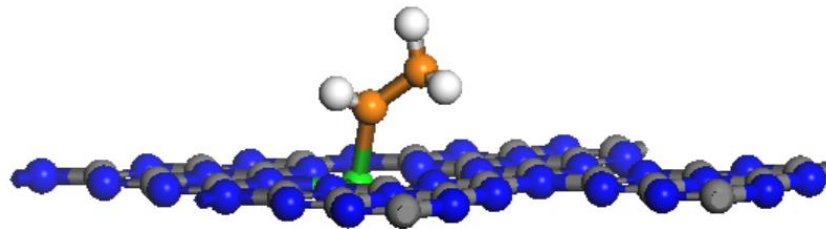
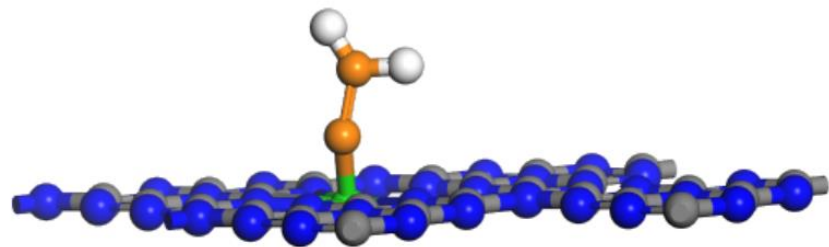


\*NHNH

Ps: 为更清楚显示, 氮气分子中的N用黄色表示。

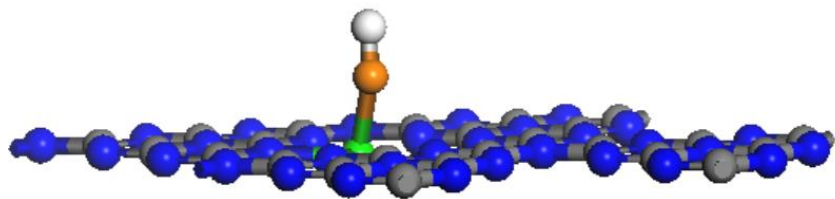


**NRR**关键中间体吸附模型（以Pd@C<sub>3</sub>N<sub>4</sub>面的NRR为例）

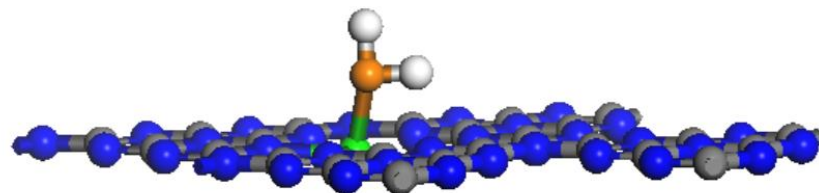


**Ps:** 为更清楚显示，氮气分子中的N用黄色表示。

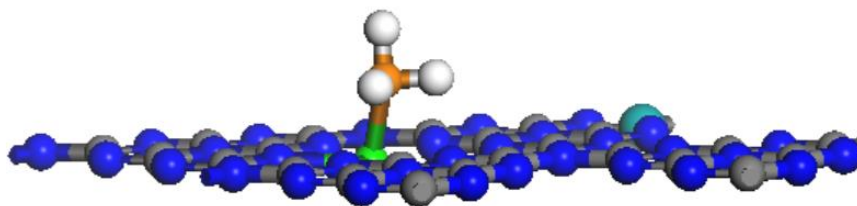
## NRR关键中间体吸附模型（以Pd@C<sub>3</sub>N<sub>4</sub>面的NRR为例）



\*NH



\*NH<sub>2</sub>



\*NH<sub>3</sub>

Ps: 为更清楚显示, 氮气分子中的N用黄色表示。

```
|SYSTEM =Fe-NH2

#Startparameter for this run:
PREC   = Normal
ISTART  = 0
ICHARG  = 2
ISPIN=2

ENCUT   = 400
NELM    = 500; #maximum of electronic step
EDIFF   = 1E-5
LREAL   = Auto
EDIFFG  = -1E-2
NSW     = 1000 # number of ioinc step
IBRION  = 2 # optimization
ISIF    = 2
ISMEAR  =0
SIGMA   = 0.2
IVDW=11
    VDW_RADIUS=50.2
    VDW_CNRADIUS=20.0
|GGA=PE|
```

提交文件与结构优化相同

可能遇到问题：

1. 指定的位点结构不存在

2. 加氢前后位点发生迁移

## 热力学第二定律：

表述1：不可能把热量从低温物体传向高温物体而不引起其它变化。

表述2：不可能制成一种循环动作的热机，从单一热源取热，使之完全变为功而不引起其它变化。。

$$\Delta G = \Delta H - T\Delta S < 0$$

熵增原理：孤立系统的熵永不自动减少，熵在可逆过程中不变，在不可逆过程中增加。。

## 标准氢电极：

17886

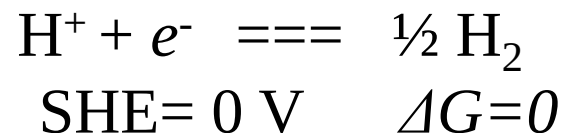
*J. Phys. Chem. B* 2004, 108, 17886–17892

### Origin of the Overpotential for Oxygen Reduction at a Fuel-Cell Cathode

J. K. Nørskov,\* J. Rossmeisl, A. Logadottir, and L. Lindqvist

*Center for Atomic-scale Materials Physics, Department of Physics, Technical University of Denmark, DK-2800 Lyngby, Denmark*

1. By setting the reference potential to be that of the standard hydrogen electrode, we can relate the chemical potential (the free energy per H) for the reaction ( $H^+ + e^-$ ) to that of  $1/2 H_2$  (eq 2 is in equilibrium). This means that, at pH = 0 in the electrolyte and 1 bar of  $H_2$  in the gas phase at 298 K, the reaction free energies of eqs 6 and 7 are equal to those of the reverse reactions eq 5 and eq 4 + 5 at an electrode potential of  $U = 0$  relative to the standard hydrogen electrode.



和频率有关

$$G = E_{DFT} + E_{ZPE} - TS + \int C_p dT$$

体系吉布斯自由能

DFT计算得到的体系能量

DFT计算得到的体系零点矫正能

温度

体系熵

体系焓



**ZPE**指的是零点能校正，就是说即使在**0K**下物质的运动也是绝对的，不可能完全静止的。我们计算能量的时候，需要一个能量为**0**的参考标准，物质不同，参考标准不同。所以需要把这个**ZPE**计算出来并扣除。计算公式如下：

$$ZPE = \sum_i \frac{1}{2} h * c * \nu_i$$

热力学概率 **$\Omega$** 定义为系统任一宏观态对应的微观态数，即概率分布数。玻耳兹曼用下面关系式定义的熵 **$S$** 和热力学概率 **$\Omega$** 联系起来，来表示系统无序性。

$$S = k \ln \Omega$$

理论计算中，热力学概率 **$\Omega$** 即为分子的频率，熵值和频率的关系如下：

$$T * S = k_B T \left\{ \frac{h * c * \nu_i}{k_B T} * \frac{1}{\exp\left(\frac{h * c * \nu_i}{k_B T}\right) - 1} - \ln\left[1 - \exp\left(-\frac{h * c * \nu_i}{k_B T}\right)\right] \right\}$$

$$G = E_{DFT} + E_{ZPE} - TS + \int C_p dT$$

第一步：结构优化

得到  $E_{DFT}$

第二步：计算频率

得到  $ZPE$ , 熵值和焓

值

因此计算  $G$  值，只要优化结构得到能量，和计算频率就可以得到

# 计算气相CO<sub>2</sub>的G值为例

## 第一步优化CO<sub>2</sub>分子

### INCAR

SYSTEM =opt

#Startparameter for this run:

PREC = Accurate

ISTART = 0

ICHARG = 2

# ISPIN = 2

ENCUT = 520

NELM = 100; #maximum of electronic step

NELMIN = 8

EDIFF = 1E-5

LREAL=Auto

EDIFFG =-1E-2

NSW = 1000 # number of ioinc step

IBRION = 2 # optimization

ISIF = 2

ISMear = 0

SIGMA = 0.2

IVDW = 11

VDW\_RADIUS = 50.2

VDW\_CNRADIUS = 20.0

### KPOINTS

Automatic mesh

0

G

1 1 1

0. 0. 0.

# 计算气相CO<sub>2</sub>的G值为例

## POSCAR

CO2

1.0000000000000000

12.5500001907000005 0.0000000000000000 0.0000000000000000

-6.2750000954000003 10.8686189826999993 0.0000000000000000

0.0000000000000000 0.0000000000000000 15.0000000000000000

C O

1 2

Direct

0.5227803420549650 0.5608565009982810 0.5030767569867285

0.5813324955247222 0.6252600863501253 0.4440003190024233

0.4641973044203074 0.4964535116515966 0.5621429280108478

## POTCAR

PAW\_PBE C 08Apr2002

4.0000000000000000

parameters from PSCTR are:

VRHFIN = C: s2p2

LEXCH = PE

EATOM = 147.1560 eV, 10.8157 Ry

TITEL = PAW\_PBE C 08Apr2002

LULTRA = F use ultrasoft PP ?

IUNSCR = 1 unscreen: 0-lin 1-nonlin 2-no

RPACOR = 1.200 partial core radius

POMASS = 12.011; ZVAL = 4.000 mass and valenz

RCORE = 1.500 outmost cutoff radius

RWIGS = 1.630; RWIGS = 0.863 wigner-seitz radius (au A)

ENMAX = 400.000; ENMIN = 300.000 eV

ICORE = 2 local potential

LCOR = T correct aug charges

LPAW = T paw PP

EAUG = 644.873

DEXC = 0.000

RMAX = 1.529 core radius for proj-oper

RAUG = 1.300 factor for augmentation sphere

RDEP = 1.501 radius for radial grids

RDEPT = 1.300 core radius for aug-charge

# 计算气相CO<sub>2</sub>的G值为例

第一步优化CO<sub>2</sub>分子后得到一系列文件：

结构文件**CONTCAR**

结果文件**OUTCAR**,得到E<sub>DFT</sub>

|  |               |             |           |
|--|---------------|-------------|-----------|
|  | CHG           | 99,916,5... | 文件        |
|  | CHGCAR        | 150,289,... | 文件        |
|  | CONTCAR       | 633         | 文件        |
|  | DOSCAR        | 11,073      | 文件        |
|  | EIGENVAL      | 784         | 文件        |
|  | IBZKPT        | 132         | 文件        |
|  | INCAR         | 901         | 文件        |
|  | KPOINTS       | 37          | 文件        |
|  | OSZICAR       | 1,948       | 文件        |
|  | OUTCAR        | 69,082      | 文件        |
|  | PCDAT         | 234         | 文件        |
|  | POSCAR        | 633         | 文件        |
|  | POTCAR        | 427,422     | 文件        |
|  | REPORT        | 0           | 文件        |
|  | sub.sh        | 461         | SH 文件     |
|  | vaspjob.o348  | 1,335       | O348 文... |
|  | vaspjob.po348 | 225         | PO348 ... |
|  | vasprun.xml   | 40,472      | XML Do... |
|  | WAVECAR       | 8,387,056   | 文件        |
|  | XDATCAR       | 330         | 文件        |

**tail -c 150000 OUTCAR**, 得到E<sub>DFT</sub>

FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

-----  
free energy TOTEN = -22.95844939 eV

energy without entropy= -22.95844939 energy(sigma>0) = -22.95844939

所以CO<sub>2</sub>的E<sub>DFT</sub> = -22.95844939 eV

# 计算气相CO<sub>2</sub>的G值为例

## 第二步计算CO<sub>2</sub>分子频率：

SYSTEM =Cu-GLY

#Startparameter for this run:

PREC = Accurate

ISTART = 0

ICHARG = 2

#ISPIN=2

ENCUT = 400

NELM = 500; #maximum of electronic step

EDIFF = 1E-4

LREAL = Auto

EDIFFG = -0.2

NSW = 1 # number of ioinc step

IBRION = 5 # optimization

NFREE = 2

POTIM = 0.015

ISIF = 2

ISMear = 0

SIGMA = 0.1

IVDW=11

GGA=PE

计算频率的**INCAR**

**KPOINTS (不变)**

Automatic mesh

0

G

1 1 1

0.0.0.



# 案例一：计算气相CO<sub>2</sub>的G值为例

将第一步结构优化的CONTCAR重命名为POSCAR

命令：mv CONTCAR POSCAR

CO2

1.0000000000000000

12.5500001907000005 0.0000000000000000 0.0000000000000000

-6.2750000954000003 10.8686189826999993 0.0000000000000000

0.0000000000000000 0.0000000000000000 15.0000000000000000

C O

1 2

Direct

0.5227803420549650 0.5608565009982810 0.5030767569867285

0.5813324955247222 0.6252600863501253 0.4440003190024233

0.4641973044203074 0.4964535116515966 0.5621429280108478

# 计算气相CO<sub>2</sub>的G值为例

## 第二步计算CO<sub>2</sub>频率得到一系列文件:

|                 |             |           |        |
|-----------------|-------------|-----------|--------|
| CHG             | 133,872,... | 文件        | 2021/0 |
| CHGCAR          | 100,682,... | 文件        | 2021/0 |
| CONTCAR         | 688         | 文件        | 2021/0 |
| DOSCAR          | 11,073      | 文件        | 2021/0 |
| DYNMAT          | 1,239       | 文件        | 2021/0 |
| EIGENVAL        | 1,312       | 文件        | 2021/0 |
| IBZKPT          | 132         | 文件        | 2021/0 |
| INCAR           | 1,080       | 文件        | 2021/0 |
| KPOINTS         | 36          | 文件        | 2021/0 |
| O2-sideon.o576  | 18,975      | O576 文... | 2021/0 |
| O2-sideon.po576 | 369         | PO576 ... | 2021/0 |
| OSZICAR         | 11,071      | 文件        | 2021/0 |
| OUTCAR          | 397,645     | 文件        | 2021/0 |
| PCDAT           | 234         | 文件        | 2021/0 |
| POSCAR          | 497         | 文件        | 2021/0 |
| POTCAR          | 427,422     | 文件        | 2021/0 |
| REPORT          | 0           | 文件        | 2021/0 |
| sub.sh          | 463         | SH 文件     | 2021/0 |
| SYMMOL.out      | 2,671       | OUT 文件    | 2021/0 |
| vasprun.xml     | 119,287     | XML Do... | 2021/0 |
| WAVECAR         | 10,421,6... | 文件        | 2021/0 |
| XDATCAR         | 2,886       | 文件        | 2021/0 |

采用vaspkit软件处理数据得到零点矫正能，熵值和焓值

# 计算气相CO<sub>2</sub>的G值为例

## 输入:vaspkit

```
[caoyy@ibmnode CO2]$ vaspkit

  \\\\/\\
  /  _  \
 (| (.) (.) |)
Hey, you must know what you are doing.
Otherwise you might get wrong results!
+-----o000o--()--o000o-----+
|                               |
|      VASPKIT Version: 1.12 (01 Mar. 2020)      |
|      Core Developer: Vei WANG (wangvei@icloud.com)      |
|      Main Contributors: Nan XU, Jin-Cheng LIU & Gang TANG      |
|      Please send Bugs and Suggestions to vaspkit@gmail.com      |
|-----o000o-----+
|
|      ( )      Oooo.
|      \ (      ( )
|      \ )      ) /
|      ( /
|
===== Structural Options =====
1)  VASP Input Files Generator      2)  Elastic-Properties
3)  K-Path Generator                4)  Structure Editor
5)  Catalysis-ElectroChem Kit       6)  Symmetry Search

===== Electronic Options =====
11) Density-of-States              21) DFT Band-Structure
23) 3D Band-Structure              25) Hybrid-DFT Band-Structure
26) Fermi-Surface                  28) Band-Structure Unfolding

===== Charge & Potential & Wavefunction Options =====
31) Charge & Spin Density          42) Potential-Related
51) Wave-Function Analysis

===== Misc Utilities =====
71) Optical-Properties              72) Molecular-Dynamics Kit
73) VASP2other Interface           74) USER interface
91) Semiconductor Calculator        92) 2D-Materials Kit

0)  Quit
```

# 计算气相CO<sub>2</sub>的G值为例

输入:5

```
5
===== Catalysis-ElectroChem Kit =====
501) Thermal Corrections for Adsorbate
502) Thermal Corrections for Gas
503) d-Band Center (experimental)
504) Convert NEB-Path to PDB Format for Animation
505) Interpolate NEB Images Linearly
507) Imaginary Frequencies Correction
508) Bader2PQR (Shown in VMD by atomic charge)
509) Evaluate half life period for a first order reaction

0)   Quit
9)   Back
----->>
```

# 计算气相CO<sub>2</sub>的G值为例

输入:502后依次输入298（温度），1（压力）

```
-->>
502
+----- Warm Tips -----+
| See An Example in vaspkit/examples/thermo_correction/ethanol. |
| This Feature Was Contributed by Nan XU, Jincheng Liu and Sobereva. |
| Included Vibrations, Translation, Rotation & Electron contributions. |
| GAS molecules should not be with any fix. |
|-->> (01) Reading Structural Parameters from CONTCAR File...
+-----+
| Selective Dynamics is Activated! |
+-----+
-->> (02) Analyzing Molecular Symmetry Information...
Molecular Symmetry is: D(inf)h
Linear molecule found!
+-----+
Please input Temperature (K) !
298
Please input Pressure (Atm) !
1
```

# 计算气相CO<sub>2</sub>的G值为例

```
-->> (02) Analyzing Molecular Symmetry Information...
Molecular Symmetry is: D(inf)h
Linear molecule found!
+-----+
Please input Temperature (K)!
98
Please input Pressure (Atm)!

Please input Spin multiplicity!--(Number of Unpaired electron + 1)
```

```
----->>
-->> (03) Extracting frequencies from OUTCAR...
-->> (04) Reading OUTCAR File...
-->> (05) Calculating Thermal Corrections...
```

```
U(T) = ZPE + Delta_U(0->T)
H(T) = U(T) + PV = ZPE + Delta_U(0->T) + PV
G(T) = H(T) - TS = ZPE + Delta_U(0->T) + PV - TS
```

```
Zero-point energy E_ZPE      :      7.073 kcal/mol      0.306689 eV
Thermal correction to U(T):      8.741 kcal/mol      0.379015 eV
Thermal correction to H(T):      9.333 kcal/mol      0.404694 eV
Thermal correction to G(T):     -5.927 kcal/mol     -0.256996 eV
Entropy S                      :      214.245 J/(mol*K)      0.002220 eV
```

Thanks to Sobereva! (sobereva@sina.com)

```
+-----+
|               * ACKNOWLEDGMENTS *               |
| Other Contributors: Xue-Fei LIU, Peng-Fei LIU, Dao-Xiong WU, |
| Zhao-Fu ZHANG, Tian WANG, Ya-Chao LIU, Qiang LI, iGo and You! |
+-----+
|               * CITATIONS *               |
| We Would Appreciate if You Cite in Your Research with VASPKIT. |
| [1] V. Wang, N. Xu, J.C. LIU, G. Tang, et al, VASPKIT: A Pre- |
| and Post-Processing Program for VASP Code, arXiv:1908.08269. |
+-----+
```

**zpe**

**$C_{pd}T = U(T) - zpe$**

**S**



# 计算 $\text{CO}_2$ 吸附在催化剂上的G值

## 第一步优化 $\text{CO}_2$ 的吸附模型分子

### 结构优化的INCAR

SYSTEM = opt

#Startparameter for this run:

PREC = Accurate

ISTART = 0

ICHARG = 2

# ISPIN = 2

ENCUT = 520

NELM = 100; #maximum of electronic step

NELMIN = 8

EDIFF = 1E-5

LREAL=Auto

EDIFFG = -1E-2

NSW = 1000 # number of ioinc step

IBRION = 2 # optimization

ISIF = 2

ISMEAR = 0

SIGMA = 0.2

IVDW = 11

VDW\_RADIUS = 50.2

VDW\_CNRRADIUS = 20.0

### KPOINTS

Automatic mesh

0

G

4 4 1

0 0 0

周期性模型K点

设置为441

## 第一步优化CO<sub>2</sub>吸附模型得到OUTCAR:

**tail -c 150000 OUTCAR, 得到E<sub>DFT</sub>**

```
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)
-----
free  energy   TOTEN  =       -217.61687130 eV

energy  without entropy=       -217.60847090   energy(sigma->0) =       -217.612671
10

d Force = 0.5918605E-04[ 0.201E-04, 0.982E-04]   d Energy = 0.5794687E-04 0.124E
-05
d Force = 0.2540678E-01[ 0.266E-01, 0.243E-01]   d Ewald  = 0.2540696E-01-0.172E
-06
```

所以CO<sub>2</sub>的E<sub>DFT</sub> = -217.6167 eV

# 计算CO<sub>2</sub>吸附在催化剂上的G值

## 第二步计算CO<sub>2</sub>吸附频率

SYSTEM =Cu-GLY

#Startparameter for this run:

PREC = Accurate

ISTART = 0

ICHARG = 2

#ISPIN=2

ENCUT = 400

NELM = 500; #maximum of electronic step

EDIFF = 1E-4

LREAL = Auto

EDIFFG = -0.2

NSW = 1 # number of ioinc step

IBRION = 5 # optimization

NFREE = 2

POTIM = 0.015

ISIF = 2

ISMear = 0

SIGMA = 0.1

IVDW=11

GGA=PE

计算频率的**INCAR**

**KPOINTS (不变)**

Automatic mesh

0

G

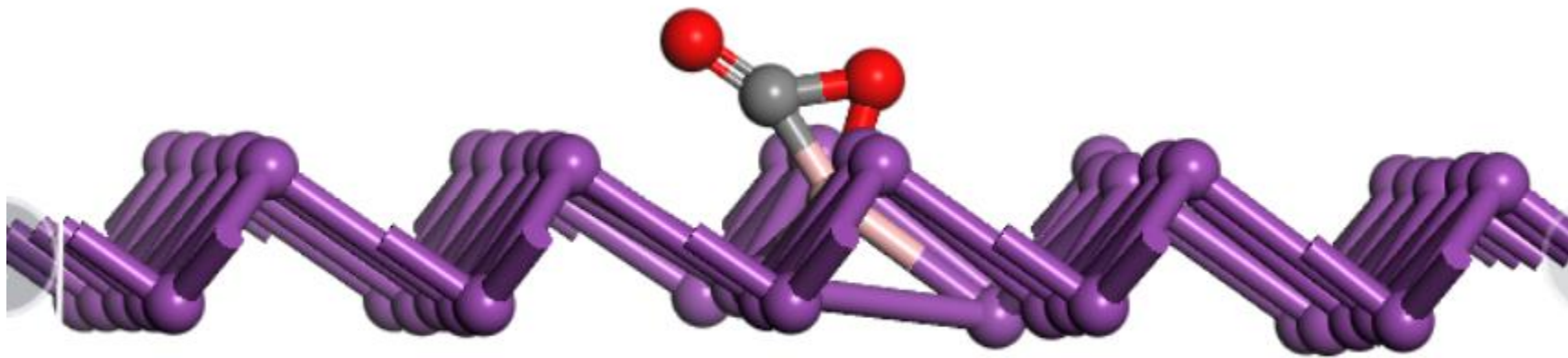
4 4 1

0 0 0

# 计算 $\text{CO}_2$ 吸附在催化剂上的G值

将第一步结构优化的**CONTCAR**重命名为**POSCAR**

命令：**mv CONTCAR POSCAR**



和气态计算频率不同的是，在计算吸附 $\text{CO}_2$ 分子的频率时，  
我们需要固定催化剂的结构。

# 计算CO<sub>2</sub>吸附在催化剂上的G值

|   |                      |                     |    |    |                          |    |    |    |
|---|----------------------|---------------------|----|----|--------------------------|----|----|----|
| 0 | 10                   | 20                  | 30 | 40 | 50                       | 60 | 70 | 80 |
| 1 | CONTCAR-B2           |                     |    |    |                          |    |    |    |
| 2 | 1.0000000000000000   |                     |    |    |                          |    |    |    |
| 3 | 21.6935005188000005  | 0.0000000000000000  |    |    | 0.0000000000000000       |    |    |    |
| 4 | -10.8712876941999994 | 18.7735119373999986 |    |    | 0.0000000000000000       |    |    |    |
| 5 | 0.0000000000000000   | 0.0000000000000000  |    |    | 21.4986991881999998      |    |    |    |
| 6 | Bi                   | B                   | C  | O  |                          |    |    |    |
| 7 | 49                   | 1                   | 1  | 2  |                          |    |    |    |
| 8 | Selective dynamics   |                     |    |    | Selective dyna           |    |    |    |
| 9 | Direct               |                     |    |    |                          |    |    |    |
| 0 | 0.075020000          | 0.011970000         |    |    | 0.338719994 F F F        |    |    | }  |
| 1 | 0.144400001          | 0.147159994         |    |    | 0.418509990 F F F        |    |    |    |
| 2 | 0.277099997          | 0.013530000         |    |    | 0.341010004 F F F        |    |    |    |
| 3 | 0.340759993          | 0.145740002         |    |    | 0.423400015 F F F        |    |    |    |
| 4 | 0.476999998          | 0.013150000         |    |    | 0.345440000 F F F        |    |    |    |
| 5 | 0.542639971          | 0.146569997         |    |    | 0.426600009 F F F        |    |    |    |
| 6 | 0.676810026          | 0.013360000         |    |    | 0.347180009 F F F        |    |    |    |
| 7 | 0.744260013          | 0.147129998         |    |    | 0.427320004 F F F        |    |    |    |
| 8 | 0.875760019          | 0.014230000         |    |    | 0.341300011 F F F        |    |    |    |
| 9 | .....                |                     |    |    |                          |    |    |    |
| 0 | 0.141479999          | 0.945179999         |    |    | 0.419140011 F F F        |    |    | }  |
| 1 | 0.276939988          | 0.813199997         |    |    | 0.341190010 F F F        |    |    |    |
| 2 | 0.341340005          | 0.945620000         |    |    | 0.423170000 F F F        |    |    |    |
| 3 | 0.477189988          | 0.811179996         |    |    | 0.346780002 F F F        |    |    |    |
| 4 | 0.542609990          | 0.945580006         |    |    | 0.426840007 F F F        |    |    |    |
| 5 | 0.675490022          | 0.815919995         |    |    | 0.344900012 F F F        |    |    |    |
| 6 | 0.746309996          | 0.948499978         |    |    | 0.426429987 F F F        |    |    |    |
| 7 | 0.870190024          | 0.807839990         |    |    | 0.337229997 F F F        |    |    |    |
| 8 | 0.939670026          | 0.942349970         |    |    | 0.418260008 F F F        |    |    |    |
| 9 | 0.545059979          | 0.547460020         |    |    | 0.404489994 F F F        |    |    |    |
| 0 | 0.4981782216243045   | 0.5243625721186984  |    |    | 0.4656741546246352 T T T |    |    | }  |
| 1 | 0.4446818496318465   | 0.4977825635354969  |    |    | 0.4972933342465569 T T T |    |    |    |
| 2 | 0.5675693444631893   | 0.5589967664158061  |    |    | 0.4759277670761145 T T T |    |    |    |

Selective dynamics 必须加

催化剂结构固定

FFF

CO<sub>2</sub>分子不固定

TTT

## 第二步计算CO<sub>2</sub>频率得到一系列文件:

|                 |             |           |        |
|-----------------|-------------|-----------|--------|
| CHG             | 133,872,... | 文件        | 2021/0 |
| CHGCAR          | 100,682,... | 文件        | 2021/0 |
| CONTCAR         | 688         | 文件        | 2021/0 |
| DOSCAR          | 11,073      | 文件        | 2021/0 |
| DYNMAT          | 1,239       | 文件        | 2021/0 |
| EIGENVAL        | 1,312       | 文件        | 2021/0 |
| IBZKPT          | 132         | 文件        | 2021/0 |
| INCAR           | 1,080       | 文件        | 2021/0 |
| KPOINTS         | 36          | 文件        | 2021/0 |
| O2-sideon.o576  | 18,975      | O576 文... | 2021/0 |
| O2-sideon.po576 | 369         | PO576 ... | 2021/0 |
| OSZICAR         | 11,071      | 文件        | 2021/0 |
| OUTCAR          | 397,645     | 文件        | 2021/0 |
| PCDAT           | 234         | 文件        | 2021/0 |
| POSCAR          | 497         | 文件        | 2021/0 |
| POTCAR          | 427,422     | 文件        | 2021/0 |
| REPORT          | 0           | 文件        | 2021/0 |
| sub.sh          | 463         | SH 文件     | 2021/0 |
| SYMMOL.out      | 2,671       | OUT 文件    | 2021/0 |
| vasprun.xml     | 119,287     | XML Do... | 2021/0 |
| WAVECAR         | 10,421,6... | 文件        | 2021/0 |
| XDATCAR         | 2,886       | 文件        | 2021/0 |

采用vaspkit软件处理数据得到零点矫正能，熵值和焓值

计算方法与气态类似

输入vaspkit---5--- 501---  
298---1



# 计算CO<sub>2</sub>吸附在催化剂上的G值

```
Temperature (K): 298.0
Zero-point energy E_ZPE      :      7.411 kcal/mol      0.321372 eV
Thermal correction to U(T):    9.308 kcal/mol      0.403612 eV
Thermal correction to H(T):    9.308 kcal/mol      0.403612 eV
Thermal correction to G(T):    5.871 kcal/mol      0.254598 eV
Entropy S                    :    48.247 J/(mol*K)      0.000500 eV
-----+-----
* ACKNOWLEDGMENTS *
Other Contributors: Xue-Fei LIU, Peng-Fei LIU, Dao-Xiong WU,
Zhao-Fu ZHANG, Tian WANG, Ya-Chao LIU, Qiang LI, iGo and You!
```

zpe

S

$$C_{pd}T = U(T) - zpe$$

最后得到将两部分能量相加得到吸附G<sub>CO<sub>2</sub></sub>的值为-217,36 eV

第一步写出**HER**基元反应步骤：

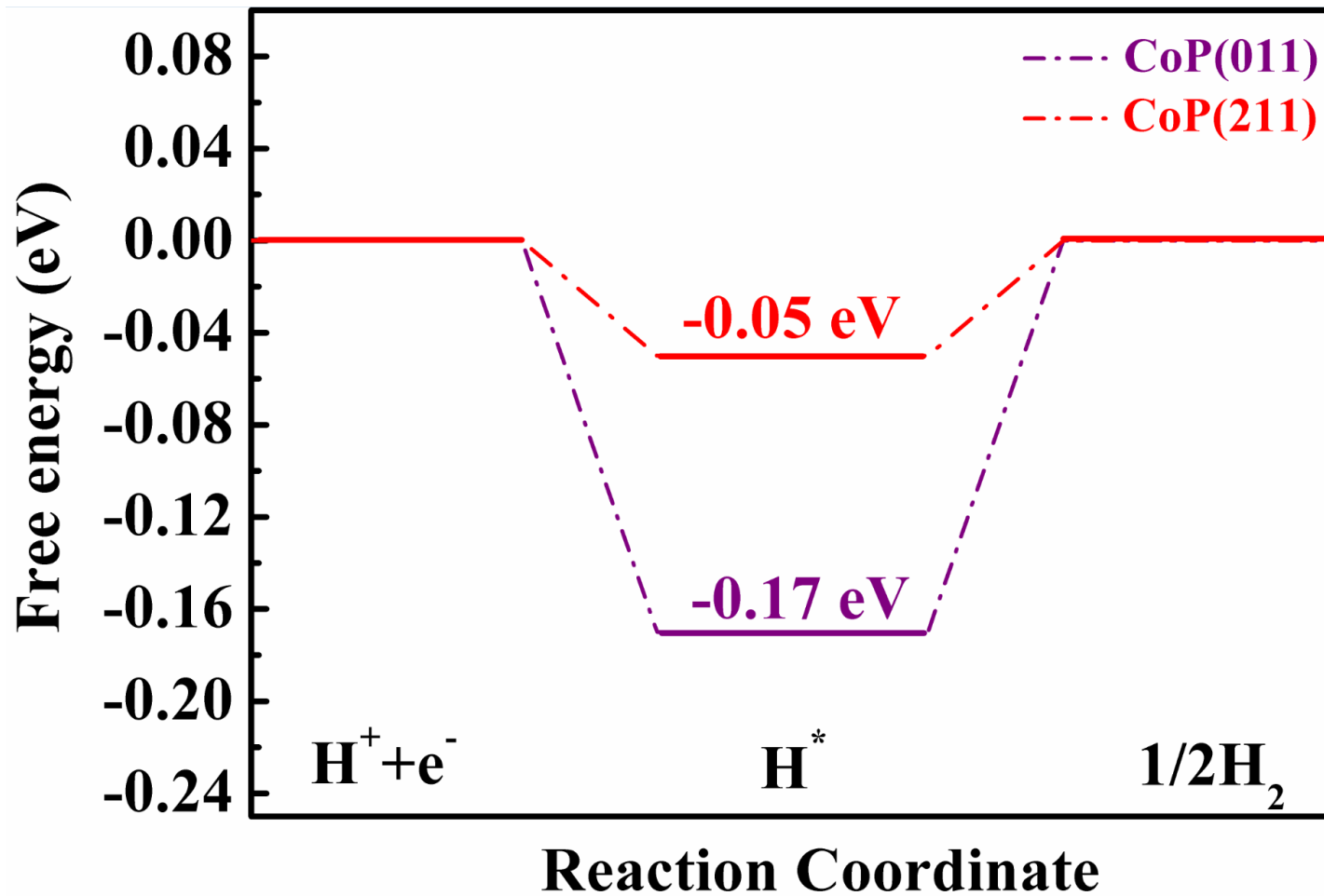


第二步计算**ORR**吸附物种的自由能：

$$\Delta G_{H^*} = \Delta E_{H^*} + \Delta E_{ZPE} - T \Delta S + \Delta \int C_p dT$$

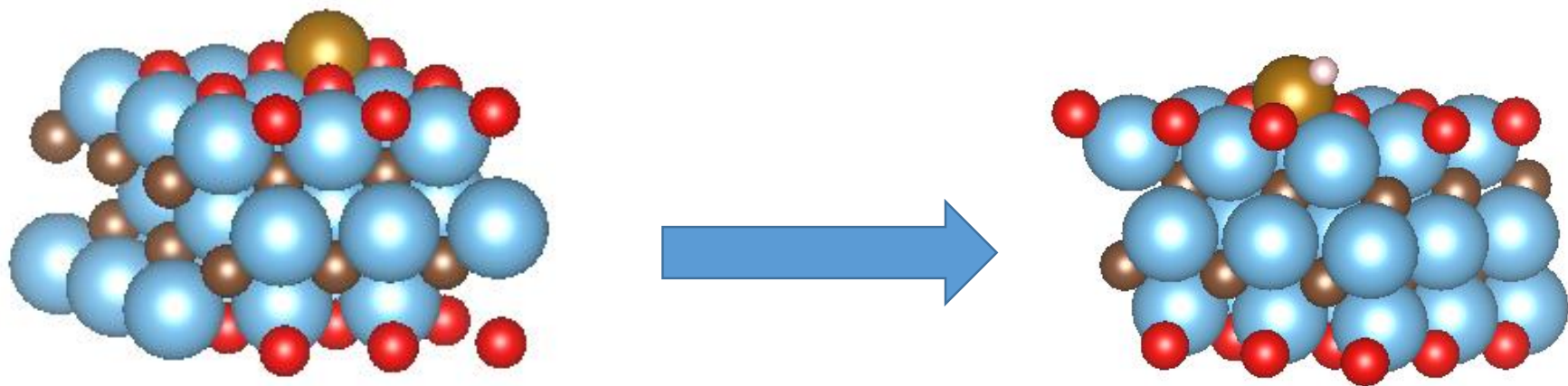
# 采用计算CHE方法计算HER台阶图

第三步画出热力学图：



$$\eta = -|\Delta G_{H^*}|/e$$

实操：计算  $* + \text{H}^+ + \text{e}^- \rightleftharpoons * \text{H}$



计算  $* + \text{H}^+ + \text{e}^- \rightleftharpoons * \text{H}$

的热力学吉布斯值变化

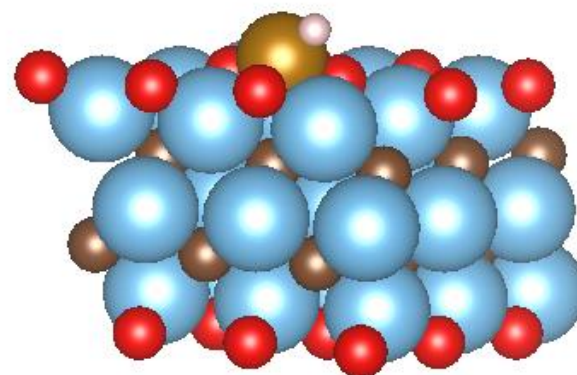
## 优化\*H结构

```
PREC = Normal
ISTART = 0
ICHARG = 2
ISPIN=2

ENCUT = 400
NELM = 500; #maximum of electronic step
EDIFF = 1E-5
LREAL = Auto
EDIFFG = -1E-2
NSW = 1000 # number of ionic step
IBRION = 2 # optimization
ISIF = 2
ISMEAR = 0
SIGMA = 0.2
IVDW=11
    VDW_RADIUS=50.2
    VDW_CNRRADIUS=20.0
# VDW_S6=14.0
# VDW_S8=16.0
# VDW_SR=1.0
GGA=PE
```

Automatic mesh

```
0
G
3 3 1
0. 0. 0.
```



优化完\*H结构，得到优化能量。

```
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)
```

```
-----  
free energy TOTEN = -586.06229325 eV
```

```
energy without entropy= -585.80360757 energy(sigma->0) = -585.93295041
```

```
d Force = 0.1114043E-07[-0.156E-06, 0.178E-06] d Energy = 0.1581517E-05-0.157E-05
```

```
d Force = 0.2729739E-02[ 0.273E-02, 0.273E-02] d Ewald = 0.2729739E-02 0.582E-10
```



优化完 **\*OOH** 结构进行频率的计算，根据频率值得出 **zpe**，熵和焓值。

**zpe，熵和焓值的计算根据上面讲过的相同方法计算。**

```

Temperature (T): 298.0 K
Zero-point energy E_ZPE      :          4.858 kcal/mol      0.210650 eV
Thermal correction to U(T):    5.005 kcal/mol      0.217041 eV
Thermal correction to H(T):    5.005 kcal/mol      0.217041 eV
Thermal correction to G(T):    4.813 kcal/mol      0.208707 eV
Entropy S                    :          2.698 J/(mol*K)    0.000028 eV/K
Entropy contribution T*S     :      804.122 J/(mol)      0.008334 eV
-----
    
```

采用 **vaspkit** 软件得出：

$$E_{ZPE}: 0.21 \text{ eV}$$

$$TS : 0.008$$

$$\int C_p dT : 0.07$$

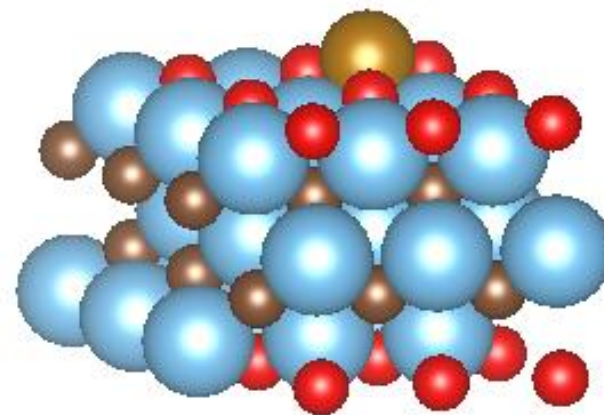
$$\begin{aligned} G &= E_{DFT} + E_{ZPE} - TS + \int C_p dT \\ &= -585.79 \text{ eV} \end{aligned}$$

## 优化催化剂结构

```
PREC  = Normal
ISTART = 0
ICHARG = 2
ISPIN=2

ENCUT  = 400
NELM   = 500; #maximum of electronic step
EDIFF  = 1E-5
LREAL  = Auto
EDIFFG = -1E-2
NSW    = 1000 # number of ionic step
IBRION = 2 # optimization
ISIF   = 2
ISMEAR = 0
SIGMA  = 0.2
IVDW=11
    VDW_RADIUS=50.2
    VDW_CNRADIUS=20.0
# VDW_S6=14.0
# VDW_S8=16.0
# VDW_SR=1.0
GGA=PE
```

```
Automatic mesh
0
G
3 3 1
0. 0. 0.
```



优化完催化剂结构，得到优化能量。

```
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)
-----
free energy TOTEN = -582.53820063 eV

energy without entropy= -582.19737759 energy(sigma->0) = -582.36778911

d Force = 0.6563859E-04[-0.212E-05, 0.133E-03] d Energy = 0.5766580E-04 0.797E-05
d Force =-0.4509275E+00[-0.451E+00,-0.450E+00] d Ewald =-0.4509276E+00 0.107E-06
```

催化剂不用计算zpe，熵和焓值。

实操：计算  $* + H^+ + e^- \text{ ----- } *H$



$$G(1/2H_2) = -3.41 \text{ eV}$$

$$G(*) = -582.24 \text{ eV}$$

$$G(*H) = -585.79 \text{ eV}$$

$$\Delta G = -585.79 - (-3.41) - (-582.24)$$

$$\Delta G = -0.41 \text{ eV}$$



$$\text{G}_{*\text{OOH}} = E_{*\text{OOH\_DFT}} + E_{*\text{OOH\_ZPE}} - TS_{*\text{OOH}} + \int Cp_{*\text{OOH\_}} dT$$

$$\text{G}_{*\text{O}_2} = E_{*\text{O}_2\_DFT} + E_{*\text{O}_2\_ZPE} - TS_{*\text{O}_2} + \int Cp_{*\text{O}_2\_} dT$$

$$\text{G}_{(\text{H}^+ + \text{e}^-)} = 1/2\text{G}(\text{H}_2) = E_{\text{H}_2\_DFT} + E_{\text{H}_2\_ZPE} - TS_{*\text{H}_2} + \int Cp_{*\text{H}_2\_} dT$$

$$\Delta\text{G1} = \text{G}_{*\text{OOH}} - \text{G}_{(\text{H}^+ + \text{e}^-)} - \text{G}_{*\text{O}_2}$$

采用相同方法可得：

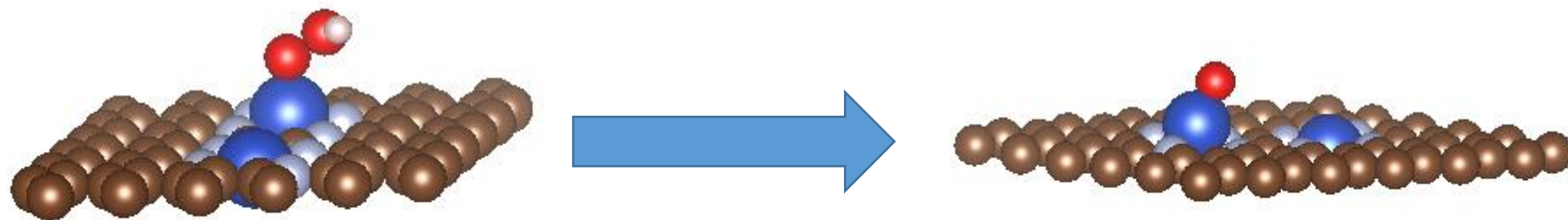
$$\Delta G2 = G_{*O} + G_{(H_2O)} - G_{(H^+ + e^-)} - G_{*OOH}$$

$$\Delta G3 = G_{*OH} - G_{(H^+ + e^-)} - G_{*O}$$

$$\Delta G4 = G^* - G_{H_2O} - G_{(H^+ + e^-)} - G_{*OH}$$



实操:  $*OOH + H^+ + e^- \rightarrow *O + H_2O$



计算  $*OOH + H^+ + e^- \rightarrow *O + H_2O$

的热力学吉布斯值变化

实操:  $*OOH + H^+ + e^- \rightarrow *O + H_2O$

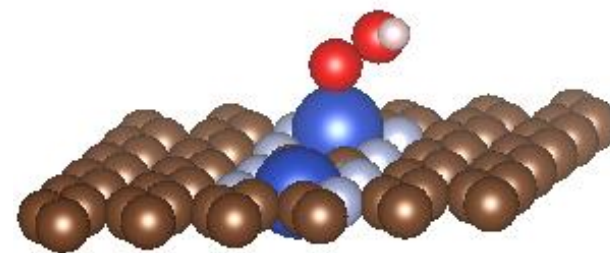
## 优化\*OOH结构

```
PREC   = Normal
ISTART = 0
ICHARG = 2
ISPIN=2

ENCUT  = 400
NELM   = 500; #maximum of electronic step
EDIFF  = 1E-5
LREAL  = Auto
EDIFFG = -1E-2
NSW    = 1000 # number of ionic step
IBRION = 2    # optimization
ISIF   = 2
ISMEAR = 0
SIGMA  = 0.2
IVDW=11
  VDW_RADIUS=50.2
  VDW_CNRRADIUS=20.0
# VDW_S6=14.0
# VDW_S8=16.0
# VDW_SR=1.0
GGA=PE
```

Automatic mesh

```
0
G
3 3 1
0. 0. 0.
```



实操:  $*OOH + H^+ + e^- \rightarrow *O + H_2O$

优化完 $*OOH$ 结构，得到优化能量。

FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

-----  
free energy TOTEN = -639.91201060 eV

energy without entropy= -639.77152957 energy(sigma->0) = -639.841770

08

实操:  $*OOH + H^+ + e^- \rightarrow *O + H_2O$

优化完 $*OOH$ 结构进行频率的计算, 根据频率值得出 $zpe$ , 熵和焓值。

$zpe$ , 熵和焓值的计算根据上面讲过的相同方法计算。

```
Temperature (T): 298.0 K
Zero-point energy E_ZPE      :      9.492 kcal/mol      0.411605 eV
Thermal correction to U(T):   11.476 kcal/mol      0.497647 eV
Thermal correction to H(T):   11.476 kcal/mol      0.497647 eV
Thermal correction to G(T):    7.180 kcal/mol      0.311339 eV
Entropy S                    :      60.322 J/(mol*K)    0.000625 eV/K
Entropy contribution T*S     :   17975.984 J/(mol)     0.186308 eV
+-----+
```

实操：  $*OOH + H^+ + e^- \rightarrow *O + H_2O$

采用 **vaspkit** 软件得出：

$$E_{ZPE}: 0.41 \text{ eV}$$

$$TS : 0.19$$

$$\int C_p dT : 0.086$$

$$\begin{aligned} G &= E_{DFT} + E_{ZPE} - TS + \int C_p dT \\ &= -639.60 \text{ eV} \end{aligned}$$

实操:  $*OOH + H^+ + e^- \rightarrow *O + H_2O$

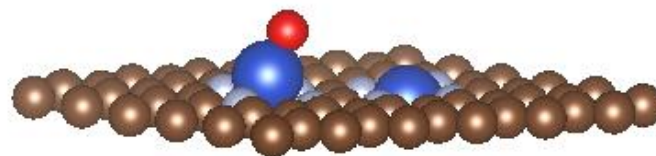
## 优化\*O结构

```
PREC = Normal
ISTART = 0
ICHARG = 2
ISPIN=2

ENCUT = 400
NELM = 500; #maximum of electronic step
EDIFF = 1E-5
LREAL = Auto
EDIFFG = -1E-2
NSW = 1000 # number of ionic step
IBRION = 2 # optimization
ISIF = 2
ISMEAR = 0
SIGMA = 0.2
IVDW=11
  VDW_RADIUS=50.2
  VDW_CNRRADIUS=20.0
# VDW_S6=14.0
# VDW_S8=16.0
# VDW_SR=1.0
GGA=PE
```

Automatic mesh

```
0
G
3 3 1
0. 0. 0.
```



实操:  $*OOH + H^+ + e^- \rightarrow *O + H_2O$

优化完\*O结构，得到优化能量。

FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

-----  
free energy TOTEN = -629.85513072 eV

energy without entropy= -629.74680326 energy(sigma->0) = -629.80096699



实操：  $*OOH + H^+ + e^- \rightarrow *O + H_2O$

优化完\*O结构进行频率的计算，根据频率值得出zpe，熵和焓值。

**zpe，熵和焓值的计算根据上面讲过的相同方法计算。**

```
Temperature (T): 298.0 K
Zero-point energy E_ZPE      :      1.022 kcal/mol      0.044339 eV
Thermal correction to U(T):    2.126 kcal/mol      0.092172 eV
Thermal correction to H(T):    2.126 kcal/mol      0.092172 eV
Thermal correction to G(T):   -0.484 kcal/mol     -0.020983 eV
Entropy S                     :      36.637 J/(mol*K)    0.000380 eV/K
Entropy contribution T*S      :    10917.874 J/(mol)     0.113156 eV
+-----+-----+
```

实操：  $*\text{OOH} + \text{H}^+ + \text{e}^- \longrightarrow * \text{O} + \text{H}_2\text{O}$

采用 **vaspkit** 软件得出：

$$E_{\text{ZPE}}: 0.04 \text{ eV}$$

$$TS : 0.11$$

$$\int C_p dT : 0.048$$

$$\begin{aligned} G &= E_{\text{DFT}} + E_{\text{ZPE}} - TS + \int C_p dT \\ &= -629.87 \text{ eV} \end{aligned}$$

实操：计算  $*OOH + H^+ + e^- \rightleftharpoons *O + H_2O$



$$G(1/2H_2) = -3.41 \text{ eV}$$

$$G(H_2O) = -13.67 \text{ eV}$$

$$G(*OOH) = -639.60 \text{ eV}$$

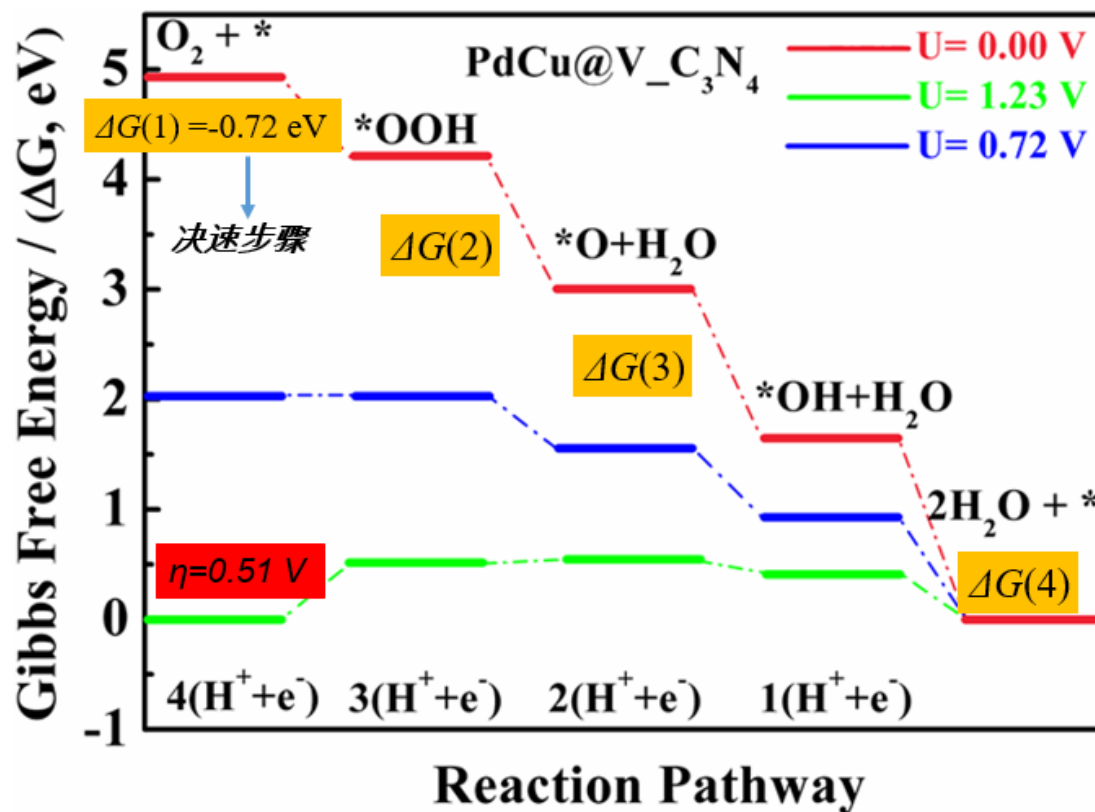
$$G(*O) = -629.87 \text{ eV}$$

$$\Delta G = -629.87 + (-13.67) - (-3.41) - (-639.60)$$

$$\Delta G = -0.53 \text{ eV}$$

实操：  $*\text{OOH} + \text{H}^+ + \text{e}^- \text{-----} * \text{O} + \text{H}_2\text{O}$

根据计算的  $\Delta G1$ ,  $\Delta G2$ ,  $\Delta G3$ ,  $\Delta G4$   
画出0电势下的台阶图，如下图的红线所示。



在外加的电势下，其质子和电子对的自由能值将发生改变，因此将导致反应的自由能值发生变化。

$$\Delta G_U = -neU$$

U表示外加的电势，n代表的是质子和电子对的数量，e代表转移电荷。

以下在限制电势和平衡电势下对**ORR**自由能的影响进行计算。

ORR限制电势  $U_{\text{limit}} = -\max\{\Delta G1, \Delta G2, \Delta G3, \Delta G4\}/e$

外加限制电势下的能垒图

$$\Delta G11 = G_{*OOH} - G_{(H^+ + e^-)} - G_{*O_2} - 4eU_{\text{limit}}$$

$$\Delta G22 = G_{*O} + G_{(H_2O)} - G_{(H^+ + e^-)} - G_{*OOH} - 3eU_{\text{limit}}$$

$$\Delta G33 = G_{*OH} - G_{(H^+ + e^-)} - G_{*O} - 2eU_{\text{limit}}$$

$$\Delta G44 = G^* - G_{H_2O} - G_{(H^+ + e^-)} - G_{*OH} - 1eU_{\text{limit}}$$

根据计算的 $\Delta G11$ ,  $\Delta G22$ ,  $\Delta G33$ ,  $\Delta G44$ 画

出外加限制电势下的台阶图，如下图的蓝线所示。

平衡电位电势下的能垒图（ORR四电子 $U_{\text{equ}}$ 为1.23 eV）

$$\Delta G_{111} = G_{*\text{OOH}} - G_{(\text{H}^+ + \text{e}^-)} - G_{*\text{O}_2} - 4eU_{\text{equ}}$$

$$\Delta G_{222} = G_{*\text{O}} + G_{(\text{H}_2\text{O})} - G_{(\text{H}^+ + \text{e}^-)} - G_{*\text{OOH}} - 3eU_{\text{equ}}$$

$$\Delta G_{333} = G_{*\text{OH}} - G_{(\text{H}^+ + \text{e}^-)} - G_{*\text{O}} - 2eU_{\text{equ}}$$

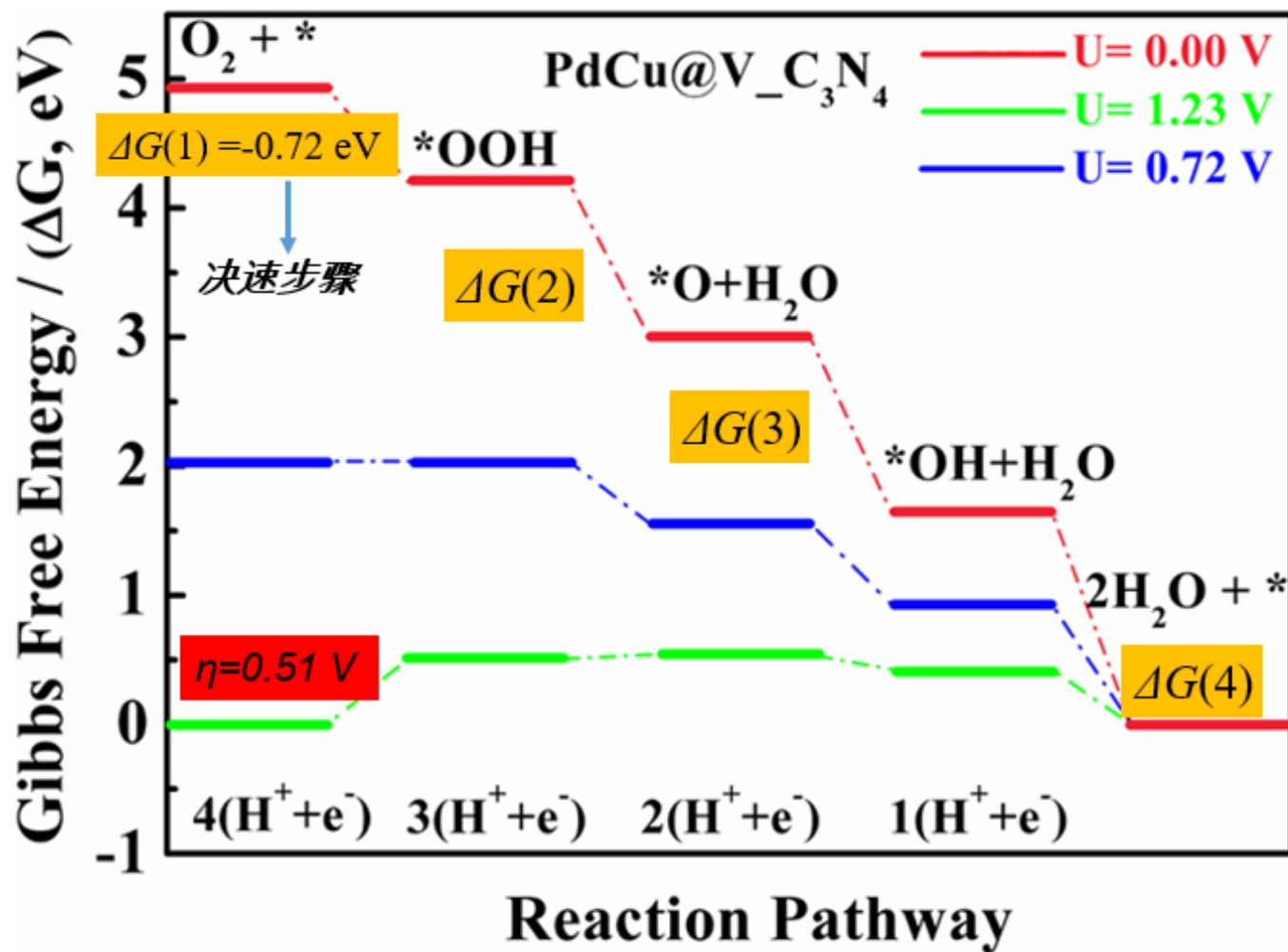
$$\Delta G_{444} = G^* - G_{\text{H}_2\text{O}} - G_{(\text{H}^+ + \text{e}^-)} - G_{*\text{OH}} - 1eU_{\text{equ}}$$

根据计算的 $\Delta G_{111}$ ,  $\Delta G_{222}$ ,  $\Delta G_{333}$ ,  $\Delta G_{444}$ 画出

外加限制电势下的台阶图，如下图的绿线所示。



# 外加电势对ORR/OER的影响



红线是0电势下；绿线是平衡电位；蓝线是外加电势下。

# 过电势 ( $\eta$ ) 计算

$$\eta = U_{equ} - U_{limit}$$

实际电极电势 ( $U$ ) = 能量下降最小的一步/ $e$

理论平衡电势 ( $U_{equ}$ ) : 根据实验得出每个反应的平衡电势, 如 $4e^-$ -ORR平衡电势1.23V.

因此上图中的过电势 :

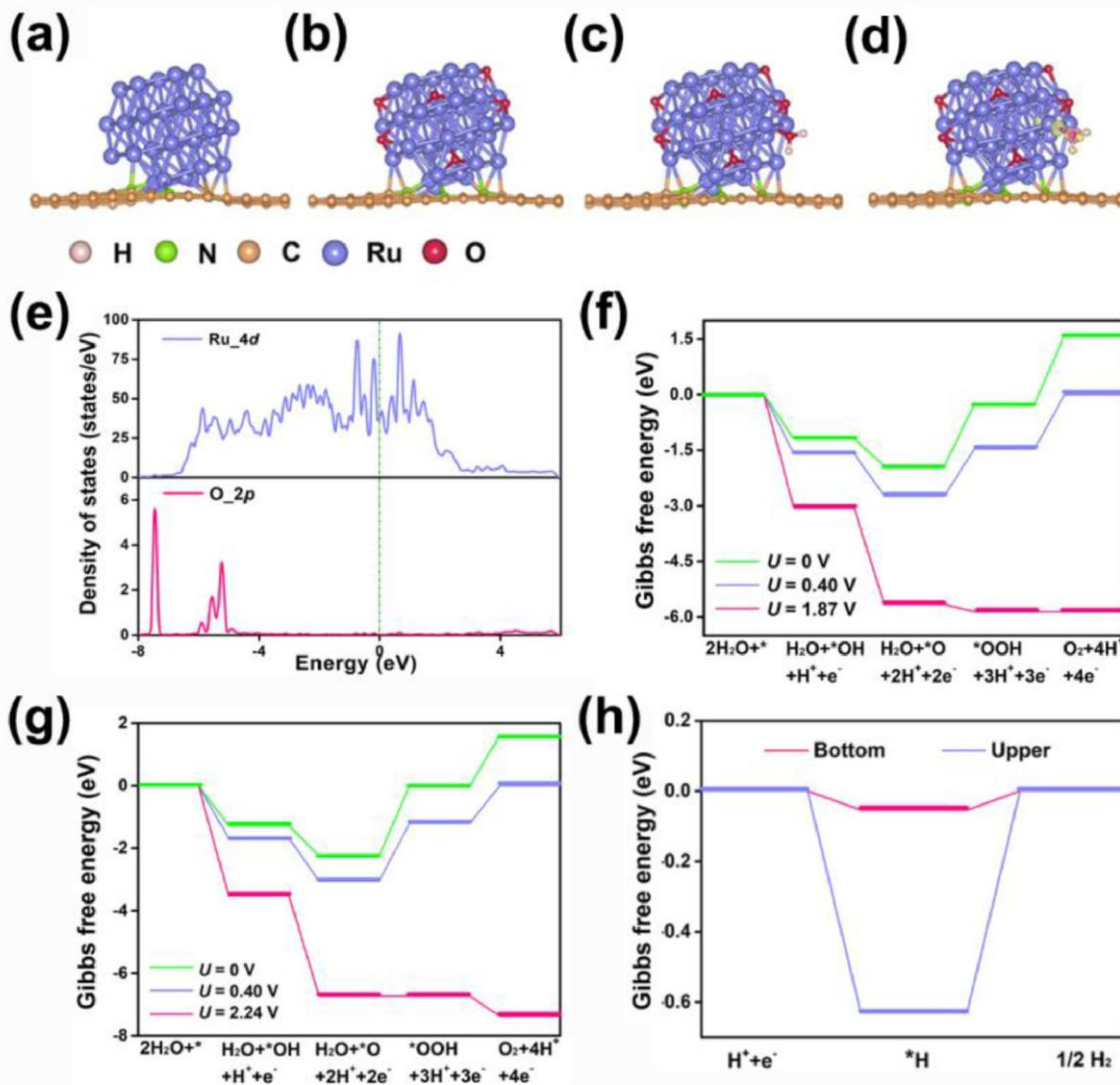
$$\eta = 1.23V - 0.72V = 0.51 V$$

在不同的酸碱性下面下，将导致反应的自由能值发生变化，其计算的公式如下：

$$\Delta G_{pH} = -k_b T \ln([H^+]) = pH * k_b T \ln(10)$$

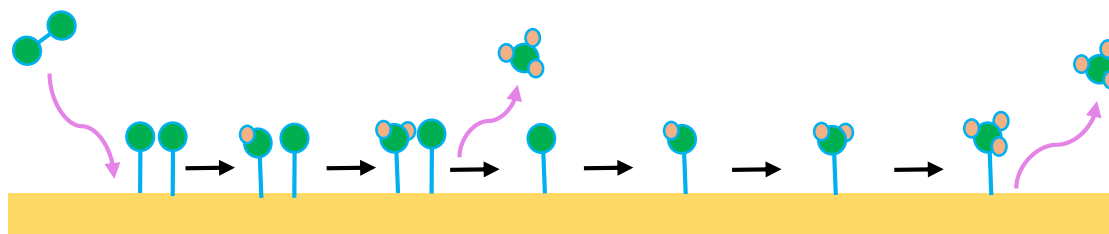
$K_b$ 是玻尔兹曼常数， $T$ 为温度

# pH值对吉布斯自由能的影响 ( $\Delta G_{\text{pH}}$ )

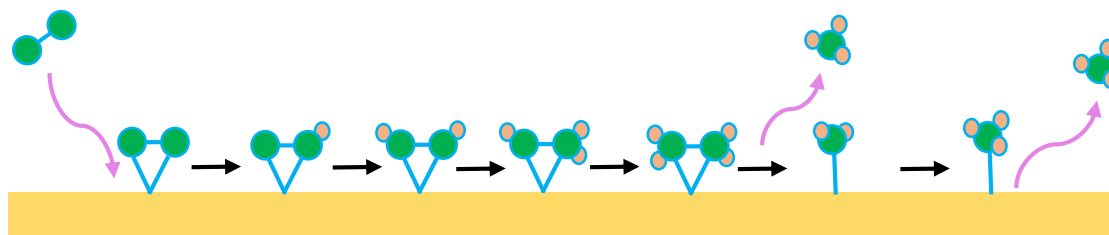


# NRR吉布斯自由能计算

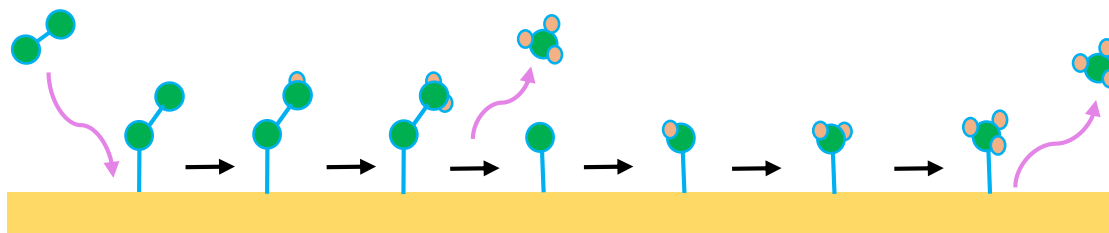
**Dissociative pathway**



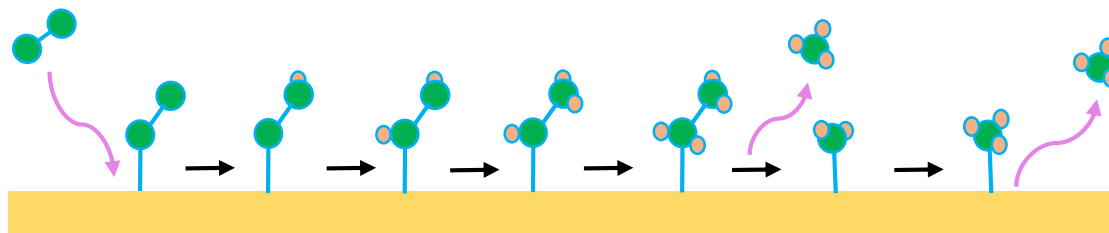
**Associative enzymatic pathway**



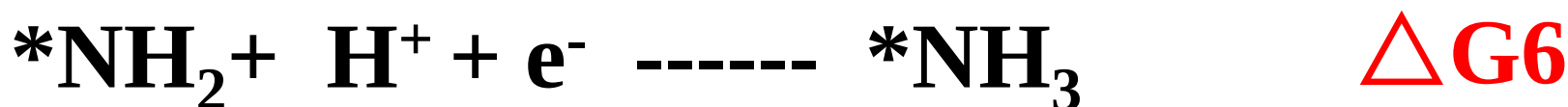
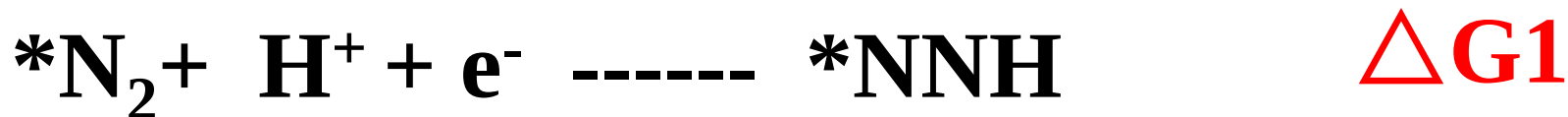
**Associative distal pathway**



**Associative alternating pathway**



以**Distal**路径计算台阶曲线为例





$$G_{*NNH} = E_{*NNH\_DFT} + E_{*NNH\_ZPE} - TS_{*NNH} + \int Cp_{*NNH} dT$$

$$G_{*N_2} = E_{*N_2\_DFT} + E_{*N_2\_ZPE} - TS_{*N_2} + \int Cp_{*N_2} dT$$

$$G_{(H^+ + e^-)} = 1/2 G(H_2) = E_{H_2\_DFT} + E_{H_2\_ZPE} - TS_{*H_2} + \int Cp_{*H_2} dT$$

$$\Delta G1 = G_{*NNH} - G_{(H^+ + e^-)} - G_{*N_2}$$

采用相同方法可得：

$$\Delta G2 = G_{*NNH2} - G_{(H+ + e-)} - G_{*NNH}$$

$$\Delta G3 = G_{*N} + G_{(NH3)} - G_{(H+ + e-)} - G_{*NNH2}$$

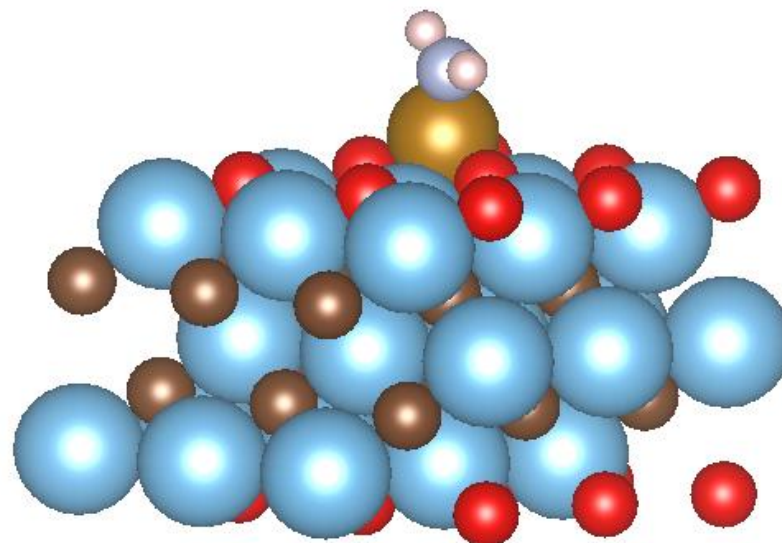
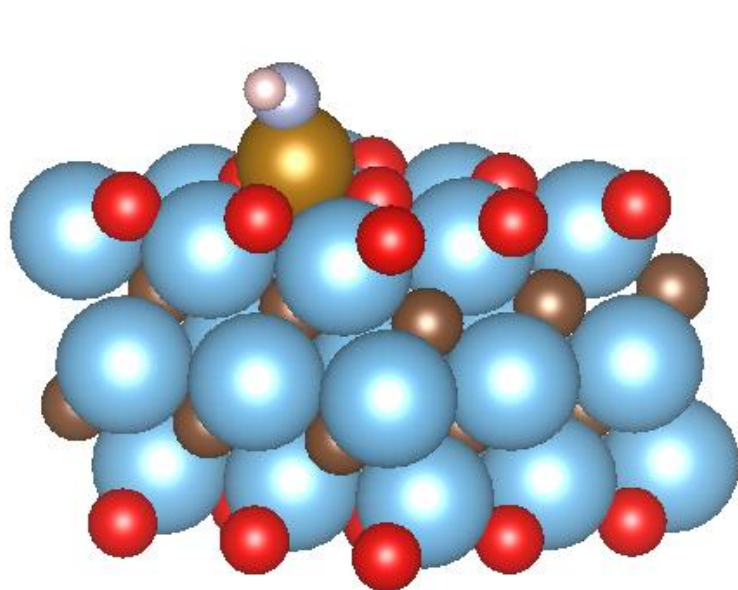
$$\Delta G4 = G_{*NH} - G_{(H+ + e-)} - G_{*N}$$

$$\Delta G5 = G_{*NH2} - G_{(H+ + e-)} - G_{*NH}$$

$$\Delta G6 = G_{*NH3} - G_{(H+ + e-)} - G_{*NH2}$$



实操： $*\text{NH} + \text{H}^+ + \text{e}^- \rightleftharpoons *\text{NH}_2$



计算 $*\text{NH} + \text{H}^+ + \text{e}^- \rightleftharpoons *\text{NH}_2$ 的热力学

吉布斯值变化

# 实操：\*NH+ H<sup>+</sup> + e<sup>-</sup> ----- \*NH<sub>2</sub>

## 优化\*NH结构

```
PREC  = Normal
ISTART = 0
ICHARG = 2
ISPIN=2

ENCUT  = 400
NELM   = 500; #maximum of electronic step
EDIFF  = 1E-5
LREAL  = Auto
EDIFFG = -1E-2
NSW    = 1000 # number of ionic step
IBRION = 2    # optimization
ISIF   = 2
ISMEAR = 0
SIGMA  = 0.2
IVDW=11
    VDW_RADIUS=50.2
    VDW_CNRADIUS=20.0
# VDW_S6=14.0
# VDW_S8=16.0
# VDW_SR=1.0
GGA=PE
```

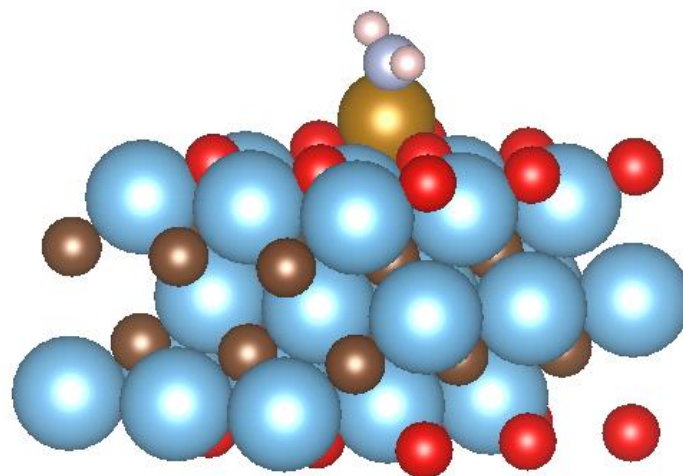
Automatic mesh

0

G

12 12 1

0. 0. 0.



实操：  $*NH + H^+ + e^- \rightarrow *NH_2$

优化完\***NH**结构，得到优化能量。

FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

-----  
free energy TOTEN = -583.776013 eV

energy without entropy= -583.539178 energy(sigma->0) = -583.657596

d Force = 0.2095317E-04[ 0.150E-04, 0.269E-04] d Energy = 0.1932910E-04 0.162E-05  
d Force = 0.3818707E+00[ 0.382E+00, 0.382E+00] d Ewald = 0.3818707E+00-0.859E-09

实操：  $*\text{NH} + \text{H}^+ + \text{e}^- \longrightarrow *\text{NH}_2$

优化完 $*\text{NH}$ 结构进行频率的计算，根据频率值得出 $\text{zpe}$ ，熵和焓值。

$\text{zpe}$ ，熵和焓值的计算根据上面讲过的相同方法计算。

```
Temperature (T): 298.0 K
Zero-point energy E_ZPE      :      15.766 kcal/mol      0.683695 eV
Thermal correction to U(T):   17.006 kcal/mol      0.737441 eV
Thermal correction to H(T):   17.006 kcal/mol      0.737441 eV
Thermal correction to G(T):   14.747 kcal/mol      0.639495 eV
Entropy S                    :      31.713 J/(mol*K)     0.000329 eV/K
Entropy contribution T*S     :    9450.385 J/(mol)       0.097946 eV
+-----+-----+
```

实操：  $*\text{NH} + \text{H}^+ + \text{e}^- \longrightarrow * \text{NH}_2$

采用 **vaspkit** 软件得出：

$$E_{\text{ZPE}}: 0.68 \text{ eV}$$

$$TS : 0.098$$

$$\int C_p dT : 0.054$$

$$\begin{aligned} G &= E_{\text{DFT}} + E_{\text{ZPE}} - TS + \int C_p dT \\ &= -583.14 \text{ eV} \end{aligned}$$

实操：  $*NH + H^+ + e^- \rightarrow *NH_2$

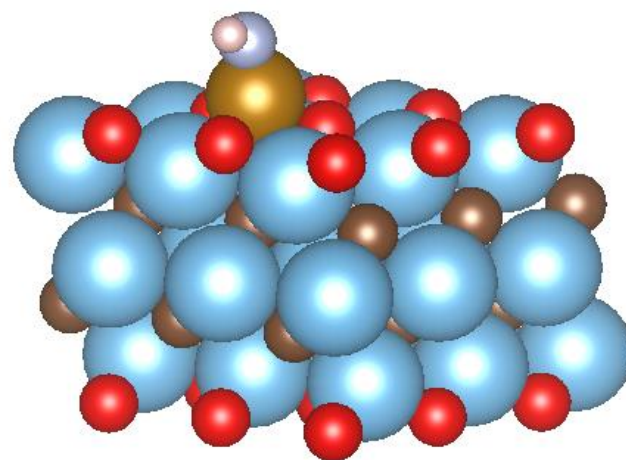
## 优化 $*NH_2$ 结构

```
PREC  = Normal
ISTART = 0
ICHARG = 2
ISPIN=2

ENCUT  = 400
NELM   = 500; #maximum of electronic step
EDIFF  = 1E-5
LREAL  = Auto
EDIFFG = -1E-2
NSW    = 1000 # number of ionic step
IBRION = 2 # optimization
ISIF   = 2
ISMEAR = 0
SIGMA  = 0.2
IVDW=11
    VDW_RADIUS=50.2
    VDW_CNRRADIUS=20.0
# VDW_S6=14.0
# VDW_S8=16.0
# VDW_SR=1.0
GGA=PE
```

Automatic mesh

```
0
G
12 12 1
0. 0. 0.
```



实操：  $*\text{NH} + \text{H}^+ + \text{e}^- \longrightarrow *\text{NH}_2$

优化完 $*\text{NH}_2$ 结构，得到优化能量。

FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

-----  
free energy TOTEN = -588.612961 eV

energy without entropy= -588.345297 energy(sigma->0) = -588.479129

实操：  $*\text{NH} + \text{H}^+ + \text{e}^- \longrightarrow *\text{NH}_2$

优化完 $*\text{NH}_2$ 结构进行频率的计算，根据频率值得出 $\text{zpe}$ ，熵和焓值。

$\text{zpe}$ ，熵和焓值的计算根据上面讲过的相同方法计算。

```
Temperature (T): 298.0 K
Zero-point energy E_ZPE      :      8.144 kcal/mol      0.353153 eV
Thermal correction to U(T):    9.123 kcal/mol      0.395614 eV
Thermal correction to H(T):    9.123 kcal/mol      0.395614 eV
Thermal correction to G(T):    7.373 kcal/mol      0.319721 eV
Entropy S                    :    24.572 J/(mol*K)    0.000255 eV/K
Entropy contribution T*S     :   7322.531 J/(mol)     0.075893 eV
```



实操：  $*\text{NH} + \text{H}^+ + \text{e}^- \longrightarrow * \text{NH}_2$

采用 **vaspkit** 软件得出：

$$E_{\text{ZPE}}: 0.35 \text{ eV}$$

$$TS : 0.076$$

$$\int C_p dT : 0.042$$

$$\begin{aligned} G &= E_{\text{DFT}} + E_{\text{ZPE}} - TS + \int C_p dT \\ &= -588.29 \text{ eV} \end{aligned}$$

实操：  $*NH + H^+ + e^- \rightleftharpoons *NH_2$



$$G(1/2H_2) = -3.41 \text{ eV}$$

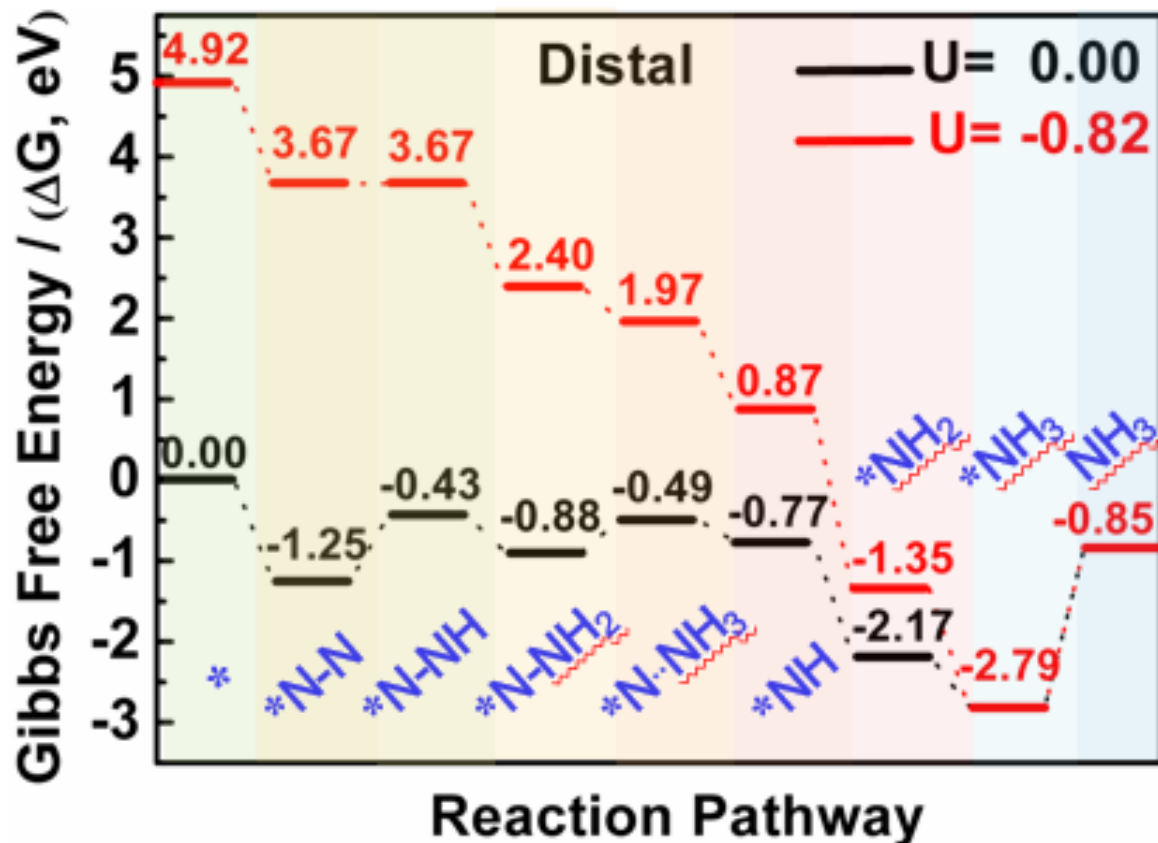
$$G(*NH) = -583.14 \text{ eV}$$

$$G(*NH_2) = -588.29 \text{ eV}$$

$$\Delta G = -588.29 - (-3.41) - (-583.14)$$

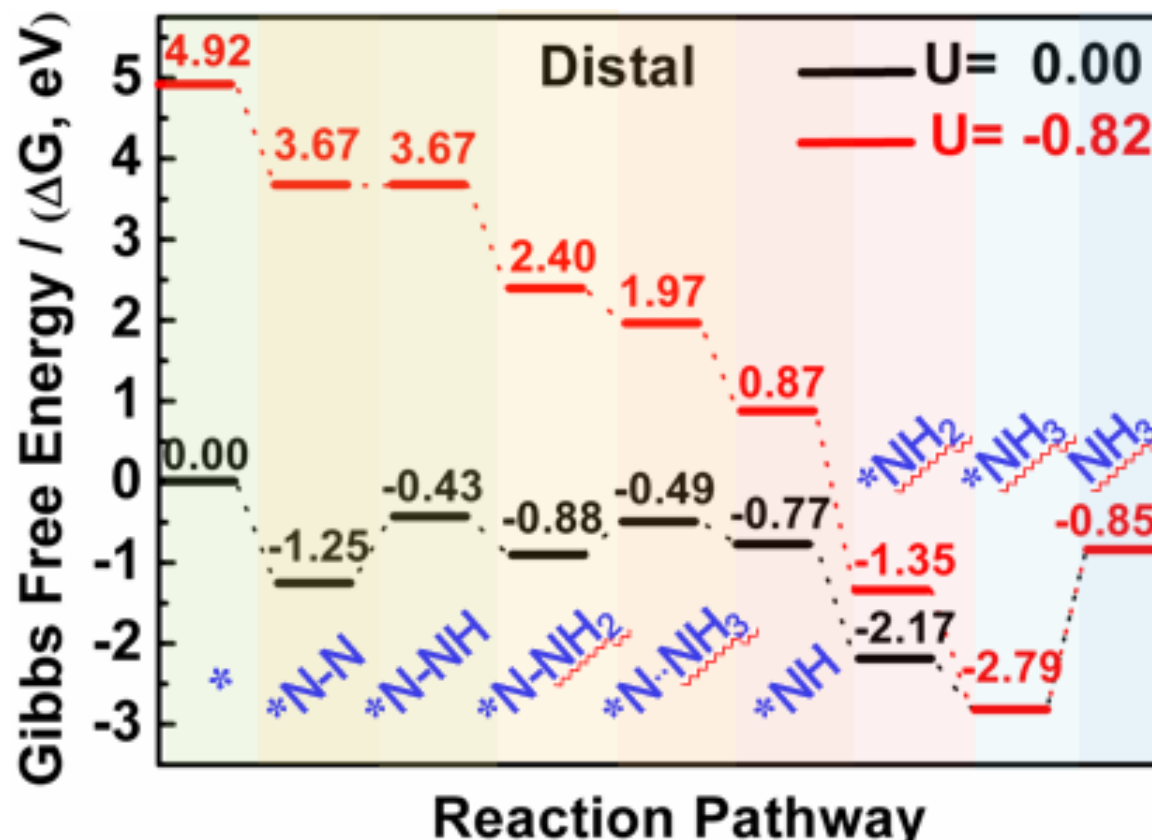
$$\Delta G = -1.75 \text{ eV}$$

## 以Distal路径计算台阶曲线为例



采用Origin画图就能得到黑色曲线

## 以Distal路径计算台阶曲线为例



$$U_{\text{limit}} = \max\{\Delta G_1, \Delta G_2, \Delta G_3, \Delta G_4, \Delta G_5, \Delta G_6\}/e^-$$

## 在外加电势下的热力学图

$$\Delta G_{11} = G_{*NNH} - G_{(H^+ + e^-)} - G_{*N_2} + 6eU_{\text{limit}}$$

$$\Delta G_{22} = G_{*NNH_2} - G_{(H^+ + e^-)} - G_{*NNH} + 5eU_{\text{limit}}$$

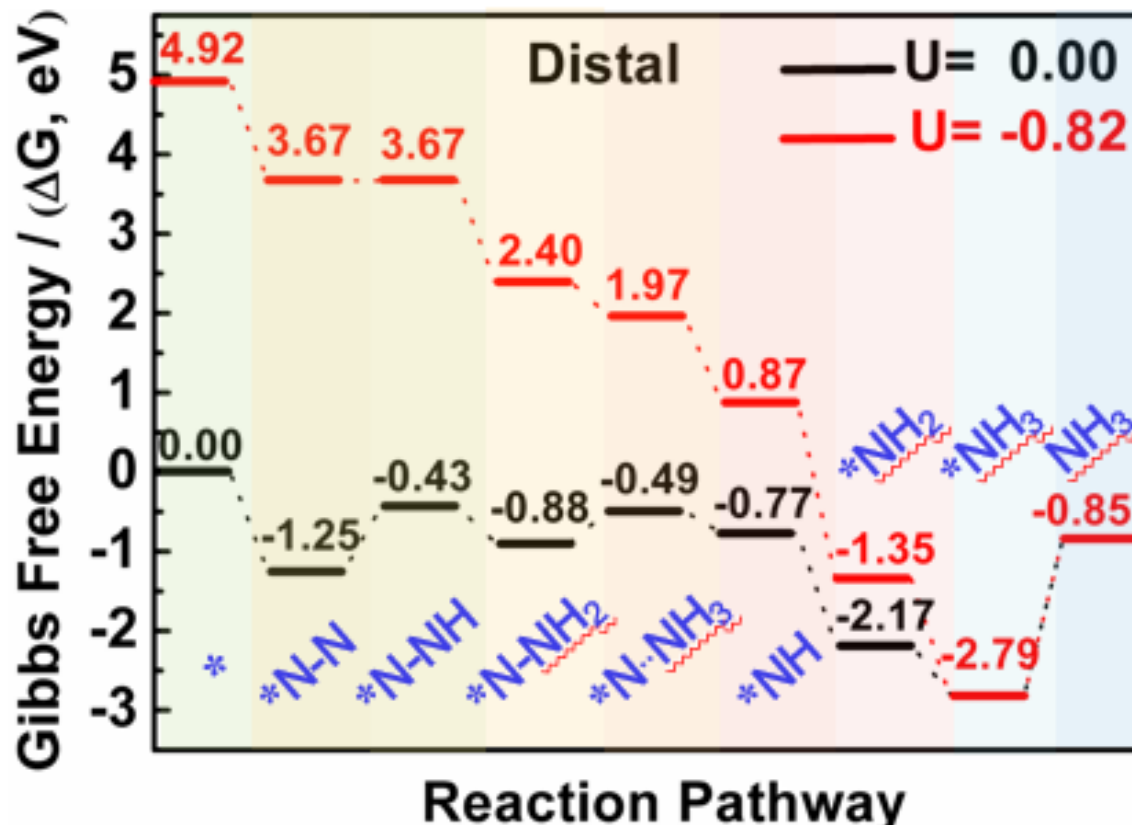
$$\Delta G_{33} = G_{*N} + G_{(NH_3)} - G_{(H^+ + e^-)} - G_{*NNH_2} + 4eU_{\text{limit}}$$

$$\Delta G_{44} = G_{*NH} - G_{(H^+ + e^-)} - G_{*N} + 3U_{\text{limit}}$$

$$\Delta G_{55} = G_{*NH_2} - G_{(H^+ + e^-)} - G_{*NH} + 2U_{\text{limit}}$$

$$\Delta G_{66} = G_{*NH_3} - G_{(H^+ + e^-)} - G_{*NH_2} + 1U_{\text{limit}}$$

## 以Distal路径计算台阶曲线为例



根据 $\Delta G_{11}$ ,  $\Delta G_{22}$ ,  $\Delta G_{33}$ ,  $\Delta G_{44}$ ,  $\Delta G_{55}$ ,  $\Delta G_{66}$ 就能画出红色的曲线

---

Thank you !